IOT-BASED FLOOD ALERT SYSTEM FOR NARVACAN, STO. TOMAS, LA UNION

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CADC
JRDC
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DEDICATION

The researchers would like to dedicate this research study to their families for their unwavering support during this research endeavor, to their teachers for their assistance and guidance in producing a final product, and to other researchers in the future who may find this study to be useful as a guide or resource.

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CADC
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ABSTRACT

DELA CRUZ, CARL ANTHONY V., DELA CRUZ, JINKY ROSE L., ESTOLERO, JUSTINE O., LACHICA, BRENDA J., PINEDA, REY JOHN E., AND VIDUYA, ANDREA P. (2024). **IoT-Based Flood Alert System for Narvacan, Sto. Tomas, La Union**. Bachelor of Science in Computer Science. Don Mariano Marcos Memorial State University South La Union Campus, College of Computer Science, Agoo, La Union.

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This study aimed to develop an IoT-based flood alert system for Narvacan, Sto. Tomas, La Union. Specifically, the study sought to achieve the following objectives: 1) develop a prototype of the IoT-Based Flood Alert System adopting the agile methodology, and 2) assess the system's level of acceptability using the User Technology Acceptability Questionnaire. Throughout the study, descriptive and developmental research methods were employed to accomplish its objectives. It focused on water level monitoring and an early warning system via SMS that alerted individuals to potential flood events, giving time to evacuate and abate the impact of flooding. The IoT-Based Flood Alert System for Narvacan, Sto. Tomas. La Union, was perceived as extremely high acceptance in terms of system usefulness, information quality, and interface quality with a grand mean of 6.87.

Keywords: *agile methodology, IoT-Based flood alert system, prototype, user technology acceptability questionnaire, water level*

Chapter 1

INTRODUCTION

Situation Analysis

Technological advancement has led the emergence of IoT-Based solutions that paved the way for a new chapter of innovation. This study embarked on a comprehensive exploration of the 'IoT-Based Flood Alert System', delving into its development, functionality, and critical role in mitigating the devastating impact of floods. The IoT-Based system comprised various components, including sensors that monitored water levels in real-time, a cloud-based platform that stored and processed the data, and a mobile application that alerted users of the current flood situation while providing them with evacuation routes and safety tips.

Cicek and Kantarci (2023) emphasized in disaster management, the massive amount of heterogeneous data that was generated through ubiquitous sensors paved the way for fast and informed decisions in the case of disasters. Utilization of the big "sensed" data led to effective and efficient management of disaster situations to prevent human and economic losses. The advancement of built-in sensing technologies in smart mobile devices enabled crowdsourcing of sensed data, which was known as mobile crowdsensing (MCS), highlighting the numerous benefits of incorporating technology into emergency response, including better tracking and locating individuals needing assistance.

The study of Shah et al. (2018) proposed a flood warning system that could detect the water level and measure the speed of the rise of water level. To give the society an earlier notification to evacuate before the water rose to the dangerous level, the

measurement results were sent as alerts to mobile phones through Short Message Service (SMS). This project was designed on the IoT-based platform, where data from the sensor was collected at the mini-processor, and alerts were generated and transmitted as SMS to smartphones.

The motivation behind the development of a flood monitoring system arose from the critical need of communities residing in flood prone areas, local disaster management agencies, or even government bodies responsible for public safety. Through surveys and discussions, it was evident that such stakeholders sought an efficient system that could provide timely flood alerts, enabling them to make informed decisions and take necessary precautions (Byali et al., 2022).

As climate-related challenges became increasingly evident, the need for effective flood monitoring and alert systems became more crucial than ever. This research paper delved into the development of flood detectors using IoT Arduino technology, aiming to provide a comprehensive solution to monitor flood situations and send timely alerts via text messages (Liang et al., 2023).

Likewise, the reports from reputable agencies consistently highlighted the urgency of improved flood monitoring and alert systems. The escalating frequency of extreme weather events, often linked to climate change, further accentuated the relevance of this contemporary issue (Radhika et al., 2022). The researchers aimed to contribute to resolving this issue by developing a practical IoT-based solution that not only identified flood situations through data analysis but also communicated the potential risks effectively, ultimately minimizing the impact of floods on lives and properties.

According to Sharma (2023), it was explored that the existing dam reservoir monitoring systems in India heavily relied on water gauges to identify potentially hazardous water levels indicative of floods. In addition, these systems faced challenges due to the remote locations of dams, resulting in high costs and significant human resources needed for continuous monitoring and maintenance. However, the primary objective of the study was to address these issues by offering a reliable, cost-effective, low-maintenance, and energy-efficient approach to intelligent flood detection and dam reservoir monitoring (Keith, 2018).

On the other hand, the advancements in computer systems revolutionized flood risk management, reducing manual efforts and enhancing coordination. With climate change and socioeconomic developments escalating flood severity, recent studies showed a transition from traditional flood control to a more adaptive and resilient approach. A comprehensive bibliometric analysis spanning from 2000 to 2021 revealed this shift in flood research trends, emphasizing a move from mitigation-focused strategies to adaptive measures and, ultimately, resilient and sustainable disaster coping plans. The integration of resilience into risk management frameworks emerged as a crucial strategy, underscoring the importance of informed decisions and proactive measures throughout the disaster management continuum (Wang et al., 2022).

The rapid pace of technological change presented challenges for users, notably in specialized fields such as coastal management and disaster preparedness. However, it concurrently engendered transformative advancements. The transition from manual, resource-intensive processes to computer-assisted monitoring and decision-making

signified a paradigm shift. The implications highlighted the pivotal role of technology in addressing intricate problems, emphasizing the paramount importance of adaptability and innovation within the realms of coastal management and disaster preparedness (Almutairi et al., 2020).

A real-time monitoring system, for instance, allowed for immediate updates on coastal conditions, allowing for swift decision-making. Using a computer, an operator could select vulnerable areas to determine if rescue efforts were necessary. This real-time data visualization assisted in efficiently identifying afflicted regions (Advanced Environmental Monitoring, 2024). These technological advancements significantly enhanced people's capacity to respond to coastal disasters, sparing lives and minimizing property loss.

In relation to this phenomenon, De Los Santos et al. (2020) designed a system leveraging a range of technologies, including mobile apps, online applications, SMS, and geolocation services, to aid individuals during crises such as fires, medical emergencies, accidents, earthquakes, floods, and storms. The primary objective of the system was to harness smartphones to enhance disaster risk mitigation and streamline incident response operations, with the ultimate goal of preserving lives.

In addition, Hoangdien (2022) developed an application called SagipPinoy that could locate the nearest emergency units, send location details to the emergency units, and broadcast an alarm to at least five closest emergency teams. The app helped regular people request help using their Android phones while also allowing emergency services to respond quickly. SagipPinoy provided a free registration service for both ordinary people and

emergency services. The researchers behind the app hoped it proved to be a secure and dependable application that could benefit emergency services and everyday people in times of need.

Similarly, Alampay et al. (2018) delved into the nature of social media and its technological affordances, shedding light on the communication needs of people at risk that were met by social media and identifying the critical functions of social media during times of crisis. However, the purpose of their study was to investigate how local government units (LGUs) had employed various information and communication technologies (ICTs), such as broadband connectivity, mobile applications, web applications, client-server network architecture, SMS, and geolocation technology, for disaster risk mitigation services. The study focused on the model local governments used to organize and manage their operations.

As stressed by Nwufoh et al. (2021), during a crisis, citizens could provide valuable support to the operation centers involved in response activities. Based on a study of relevant existent contributions in this area and their personal direct experience in developing and evaluating emergency management solutions, one of the core objectives was to provide secure and comfortable learning environments. It was essential to have better emergency alarm systems in place.

Goyal (2024) study discussed that floods were a common natural disaster that occurred almost every year in our country. Floods not only destroyed agricultural fields and produce, but they also caused significant damage to vast stretches of area and property. This was why early flood detection was extremely vital to prevent the loss of life and

valuable assets. The IoT-based flood detection system was built to monitor and track different natural factors (humidity, temperature, water level, etc.) to predict a flood, thereby allowing us to take the necessary measures to minimize the damage caused.

Flooding was considered one of the most destructive natural catastrophes that could cause serious harm to people, property, and infrastructure. According to Technology (2022), early warning systems were increasingly essential to reduce flood damage. One creative idea that offered early warning of impending floods and real-time monitoring was an Arduino-based system. These systems mainly identified floods and notified local authorities and communities to reduce damage and save lives. The sensor was an essential part of an Arduino-based system that measured water level precisely using ultrasonic technology.

By using this technology, people could help prevent the loss of life and property caused by flooding. Early warning systems like this could give people the time to evacuate and move to safety. They also allowed local authorities to make necessary preparations and take action to minimize the damage caused by floods. Overall, using an Arduino-based system with an ultrasonic sensor was an innovative and effective way to monitor water levels and provide early warning of impending floods, ultimately leading to a safer and more secure environment for everyone.

A study conducted by Gacu et al. (2022) highlighted the benefits of using mobile computing in emergency response scenarios, such as real-time tracking of individuals in need of assistance. The study found that utilizing mobile devices could significantly improve emergency response times as it allowed for accurate and timely monitoring of

individuals in need of help.

According to Wang et al. (2020), they examined the assessment of the pros and cons of various emergency response systems and suggested potential avenues for future research in this area. Their primary conclusion highlighted the importance of utilizing technology in emergency response situations to enhance the efficiency and promptness of identifying and tracking individuals in real-time using mobile phone data. The results underscored that integrating technology into emergency response systems could help conserve vital time and resources, which were particularly crucial during life-threatening emergencies. Similarly, McDermid (2022) delved into the advantages and limitations of diverse emergency response systems and proposed potential future research trajectories. The findings echoed the notion that employing technology in emergency response scenarios could augment the precision and swiftness of locating individuals in real-time, a critical aspect in urgent situations. The study suggested that the incorporation of technology into emergency response systems had the potential to preserve essential time and resources.

Additionally, the study highlighted that technology could improve user experience and safety by providing accurate information about disaster warnings via their devices. This real-time information could help individuals make informed decisions and take necessary precautions to protect themselves and others. Overall, Wang's (2020) study emphasizes the importance of technology in emergency response scenarios and its potential to enhance the effectiveness and efficiency of emergency response systems with the use of mobile phone data.

Shi et al. (2023) discussed the preventive allocation and post-disaster cooperative

dispatch of emergency mobile resources for improved distribution system resilience. The study focused on the distribution system's resilience against extreme weather events in both the preventive and post-disaster stages. The resource allocation problem was described as a two-stage stochastic programming problem in the preventative stage to minimize the anticipated load-shedding cost right after the disaster, taking the uncertainty of the fault site into account.

Furthermore, Zhang et al. (2023) studied coastal emergency response management and developed an integrated framework that included a decision support system, a risk assessment model, and a simulation system. The framework utilized various data sources, such as satellite imagery, GIS, and remote sensing, to provide decision support for emergency managers. The authors aimed to provide a comprehensive approach to coastal emergency response planning that integrated data from multiple sources and provided real-time decision support during emergencies. Likewise, Kim and Hastak (2018) explained the development of a coastal disaster information management system based on GIS that integrated various data sources, such as coastal monitoring data, weather forecasts, and disaster reports, providing real-time information to emergency managers and supporting decision-making during coastal disasters. The authors aimed to develop an integrated system to monitor coastal conditions in real-time and provide decision support to emergency managers during coastal disasters.

The need for a storm monitoring system is also stressed in the study of Moradi et al. (2020) where they proposed integrating Geographic Information System and system dynamics to assess the impacts of storm damage on coastal communities. A mobile-based

emergency response system enhanced the effectiveness of emergency response teams. The authors highlighted the benefits of using technology in emergency response scenarios, such as accurate and real-time tracking of individuals. Wang et al. (2020) discussed the development of a real-time emergency response system that utilized mobile devices. They highlighted the benefits of using technology for emergency response, such as real-time tracking and locating of individuals.

Hu et al. (2022) implemented a system with open specifications to facilitate interoperability. The typhoons and floods in Hainan Province, China, were typical scenarios to verify the system's applicability and effectiveness. The system improved the automation level of the natural disaster emergency response service and provided technical support for regional remote-sensing-based disaster mitigation in China.

Zhang et al. (2023) published a study in the Journal of Hazardous Materials on a system that integrated several components, including a geographic information system (GIS), a meteorological model, a dispersion model, and a decision support system (DSS). One possible approach to addressing these challenges involved the development of a comprehensive platform that leveraged state-of-the-art technologies to monitor coastal conditions continuously, forecast emerging hazards, and facilitate efficient communication among various stakeholders. By completing these essential gaps, the advocates envisioned a prospective scenario wherein coastal communities were enhanced in their preparedness, resources were distributed more efficiently, and protection of lives and property was ensured against the severe consequences of coastal disasters (Glupker, 2023). Aside from this, the Flood Alert System aimed to substantially contribute to improving resilience and

disaster response capacities in coastal areas susceptible to vulnerability.

In 2022, Rankin conducted a thorough study to address coastal storm risks along the Atlantic Ocean shoreline in Miami-Dade County, Florida. The study, with Miami-Dade County as the non-federal sponsor, focused on vulnerabilities related to erosion, flooding, and wave attacks during coastal storms. While concentrating solely on the Atlantic Ocean shoreline, the analysis strived to provide an alternative plan that not only mitigated risks but also identified opportunities for recreation and habitat preservation. The study acknowledged the region's high vulnerability to sea level change, anticipating its exacerbation of existing damaging forces. However, this collaborative Coastal Storm Risk Management (CSRM) initiative underscored the importance of a comprehensive strategy to safeguard structures and infrastructure, balancing immediate risk mitigation with long-term adaptation to evolving challenges, especially in the context of climate change and sea level rise (Almar et al., 2021).

Nevertheless, in formulating the feasibility study plan, meticulous attention was devoted to developing a comprehensive strategy that integrated both structural and nonstructural measures to mitigate storm damage risks within the study areas. As part of this comprehensive plan, the Coastal Storm Risk Management measures not only addressed identified problems but also strategically capitalized on opportunities outlined in the primary report. In the specific context of the Rhode Island Coastline, this Coastal Storm Risk Management General Investigation was dedicated to achieving the twin objectives of presenting a Total Project Cost, encompassing both construction and non-construction expenses, for the Tentatively Selected Plan (TSP) at the current price level. This financial

assessment was pivotal for project justification and authorization, including future cost projections for effective budgeting (Nae, 2022). Moreover, in integrating cutting-edge solutions, such as an IoT-Based flood alert system, into the nonstructural measures, the cost estimation process considered the implementation and maintenance expenses associated with this technology.

Conducted by Imani et al. (2021), the study provided a comprehensive examination of the capability of the least resilient cities in Taiwan, including Tainan, Kaohsiung, Chiayi, Taoyuan, New Taipei, and Yunlin. The research revealed diverse coping strategies employed by residents, with Kaohsiung prioritizing structural and technological approaches, while Yunlin relied more on behavioral and spiritual mechanisms. The economic vulnerabilities in Yunlin, compounded by factors such as an aging farming population and long-term labor outmigration, underscored the region's susceptibility to climate change. The analysis incorporated the impact of single and multiple disasters on coping strategies, highlighting an increased reliance on financial, behavioral, and structural approaches with the prolonged duration and heightened frequency of disasters. Notably, the study proposed a comprehensive conceptual framework for disaster management termed the 4R package Resource, Reason, Roadmap, and Respond aligning with the interconnectedness of resilience, capability, and effective disaster response (Huemer & Wang, 2021). The analysis suggested prioritizing flood mitigation measures in vulnerable regions and underscored the need for further research incorporating social aspects and higher spatial resolution for accurate risk assessment and management strategies.

This study focused on addressing the significant impact of flooding, a major

disaster disrupting the normal lifestyle of communities. In accordance with Mamat et al. (2021), recognizing the imperative for comprehensive environmental analysis, the research adopted a forward-looking approach by leveraging Internet of Things (IoT) technology and embedded systems. Specifically, the flood alerting system integrated IoT and Wireless Sensor Network (WSN) to facilitate real-time calculations, computational processing, prediction, and analysis. Employing the Waterfall model algorithm as the methodology, the study utilized Arduino uno connected to water flow sensors for water level analysis and temperature sensors for predicting humidity levels. The collected data was then processed through Arduino, programmed in Java and C++, to generate alert messages (Bhatt, 2023). The seamless integration of IoT in this real-time prediction and alerting system demonstrated its effectiveness in enhancing risk reduction by providing timely warnings to the surrounding areas. This research contributed to understanding the practical implementation of IoT in flood prediction and response, showcasing its potential impact on reducing response times and improving overall disaster preparedness by Nandhini et al. (2021).

Siddique and Qadri's (2018) study addressed the critical issue of flood control in many cities of Pakistan, where water, while essential, posed a life-threatening risk, particularly for communities residing near riverbanks. The primary focus of the research was to propose a solution using Wireless Sensor Network (WSN) technology to monitor and warn nearby villages about impending flood situations, aiming to prioritize the safety of residents. The key features of the study included the introduction of four nodes strategically placed at different river locations, constantly monitoring river water flow and

level. These nodes communicated vital information to a master node, which aggregated the data and triggered warnings in the event of a flood. The significance of this research lay in its practical approach to leveraging technology, specifically WSNs, to address the challenges posed by poor flood management and the absence of dams. By providing a systematic and technologically driven solution, the study contributed to the potential mitigation of flood-related risks and underscored the vital role technology could play in safeguarding lives and properties in flood-prone areas by Muaz (2021).

The study by Guo et al. (2021) addressed a critical challenge in geohazard assessment by focusing on mountainous regions characterized by intricate topography and dense vegetation cover. Geohazard recognition and inventory mapping were pivotal components for the creation of reliable susceptibility and hazard maps, yet their effective implementation in such challenging terrains had been limited. The research underscored the potential breakthrough offered by Light Detection and Ranging (LiDAR) technology. The utilization of airborne LiDAR data presented a novel approach to overcoming the complexities associated with mountainous landscapes. By employing LiDAR technology, the study aimed to enhance the accuracy and efficiency of geohazard recognition and inventory mapping, ultimately contributing to more robust hazard assessments in regions where traditional methods faced limitations (Bantounos et al., 2022). This research signified a noteworthy advancement in the field, leveraging technological innovation to address the unique challenges posed by complex topography and vegetation in mountainous areas.

In parallel efforts, the DOST-Philippine Institute of Volcanology and Seismology

(PHIVOLCS) had taken significant strides in geohazard recognition and inventory mapping through its participation in the multi-agency GeoRisk Philippines Initiative. A pivotal aspect of this initiative was the introduction of the HazardHunterPH mobile application tailored for Android users. Notably, the application expanded its utility beyond geological hazards to encompass weather-related threats, including floods, strong winds, and storm surges, as evidenced by Philippines (2018). This comprehensive approach not only enhanced the accessibility of hazard information but also reflected a strategic integration of geohazard recognition and inventory mapping into a user-friendly, technology-driven platform, thereby fostering a more informed and resilient community response to diverse environmental risks.

The study by Valenzuela et al. (2020) revealed a discrepancy between perceived low risk and actual vulnerability in coastal communities, particularly exemplified by Puerto Princesa City, Palawan, Philippines. Despite the existence of an extensive disaster risk database and risk maps, residents in seemingly low-risk zones may have harbored a false sense of safety. The historical storm tracks and tsunami data from IBTrACS and the United States' NGDC/WDS Global Historical Tsunami Database challenged this perception, indicating Palawan's exposure to storms and potential risks from nearby tsunami sources. This incongruity underscored the need for a nuanced and adaptive risk assessment approach that considered dynamic factors and incorporated targeted community education initiatives. The study's findings emphasized the imperative for proactive disaster management strategies, urging a reevaluation of risk criteria to ensure the resilience of coastal communities in the face of evolving natural hazards. The study of Rubio et al.

(2020) focused on index-based flood risk assessment in Metro Manila, the capital region of the Philippines and the most densely populated region in the country. Its objective was to properly address urban characteristics in flood risk assessment by introducing a specific urban-type set of physical, social, economic, and ecological indicators.

The escalating frequency of emergency and disaster situations globally, particularly in highly vulnerable nations like the Philippines, underscored the critical need for effective response mechanisms. Despite this urgency, there was a noticeable gap in the literature regarding the representation of emergency and disaster response applications on smartphones in the context of the Philippines. To address this gap, the study conducted by Orong and Hernandez (2019) endeavored to assess the acceptance level of recently launched smartphone applications designed for emergency and disaster response among citizens. Employing the Unified Theory of Acceptance and Use of Technology model as a framework, the survey aimed to delve into the factors influencing the adoption of these applications. This research not only contributed to the understanding of technology acceptance in the specific domain of emergency response but also held the potential to enhance the effectiveness of such applications, fostering a more resilient and responsive community in the face of emergencies in the Philippines.

The escalation of population density and consequent urban expansion into hazard-prone regions heightened the vulnerability of settlements. This increased exposure was compounded by a notable deficiency in pre-impact awareness among residents in these communities, potentially leading to an underestimation of the severity and impact of potential hazards. It was observed that hazard knowledge disseminated through media

channels might prove inadequate, as it often failed to directly influence comprehensive planning and evacuation strategies (World Bank Group, 2023). Furthermore, individuals lacking prior disaster experience might have misconstrued the risks they faced, particularly when confronted with hazards characterized by return periods surpassing an individual's lifetime. This underscored the imperative for targeted awareness campaigns and educational initiatives to bridge the informational gap between hazard exposure and community preparedness, constituting a pivotal aspect of comprehensive disaster risk reduction strategies (Dimitrova & Snair, 2024).

Floods are undoubtedly one of the most devastating natural calamities that ravage millions worldwide yearly. The extent of the damage they could cause ranges from the destruction of infrastructure and property to the loss of human lives and agriculture. Due to its geographic location, climate, and topography, the Philippines is especially prone to floods. In La Union the areas that are highly susceptible to flooding include Barangay Narvacan, Sto. Tomas, La Union, a coastal municipality in the northern part of Luzon Island. The inhabitants of this region faced constant threats from heavy rainfall, storm surges, and river overflow during the rainy season by Muaz (2021).

During an interview, Barangay Captain Ricardo Collado of Narvacan, Sto. Tomas, La Union, expressed concern over the recurring high tide events that occurred quarterly in the area. These high-tide incidents required prompt action to mitigate potential risks to coastal communities. He also discussed the impact of high tides on the community biannually, with a seasonal occurrence in the months of February to April and from September to December. Furthermore, weekly instances of high tides extended over three

nights and encompassed at least two days. It was imperative to address these patterns comprehensively to ensure the resilience and safety of the community (Collado, 2023).

However, the barangay was intricately divided into seven sectors, and recent data unveiled a concerning trend. Sectors five to seven, inhabited by 233 individuals, grappled with the impact of severe high tides, while sectors one to four, housing 338 out of the total 571 residents, faced moderately high tides. The tangible repercussions on the affected population's lives and livelihoods underscored the urgent need for intervention. Delving into the broader context, the Municipality of Santo Tomas had a historical backdrop of contending with the challenges posed by tidal fluctuations, a recurrent phenomenon in this coastal locale. The residents of Narvacan, Sto. Tomas, La Union had traditionally relied on manual monitoring techniques, encompassing recording high tide date and time, predicting water levels, measuring water levels, and discerning the conclusion of high tide. Additionally, they employed a calendar-based method to track tidal cycles and a manual process to monitor rising water levels. The vulnerability of the barangay to flooding was accentuated by its low-lying topography and close proximity to rivers and the sea (Collado, 2023). In light of these complexities, implementing strategic measures became paramount to mitigate the impacts, ensuring the protection and resilience of the community against imminent harm.

Due to the aforementioned scenario and problems encountered by the residents, the researchers developed the IoT-Based Flood Alert System for Narvacan, Sto. Tomas, La Union. To avert such situations, it was important to monitor and receive timely alerts about the water level situation. The system would assist in evaluating emergency preparedness

and trace the affected households in the focus area from the seven sectors of the said barangay. The function of this system provided an alert message through SMS informing the residents of the status of water level in the vicinity and to evacuate their place if necessary.

The developed IoT-Based Flood Alert System stood as a comprehensive solution, offering real-time data on water levels. This innovative system went beyond conventional monitoring technologies by incorporating features like text messages and calls, establishing a robust alert system that significantly enhanced coastal risk management and emergency response. The prototype, equipped with advanced sensors and integrated with weather stations, served as a proactive tool for tracking water levels, tides, and weather conditions. The system's core functionality lay in its ability to deliver timely alerts and warnings to 30 participants, ensuring heightened preparedness in the face of high tide or flood events. The implementation of this study not only addressed the immediate needs of the Municipality of Santo Tomas but also envisioned extending its benefits to other flood-prone areas in the Philippines, particularly empowering local communities in the province to mitigate the impact of coastal risks through effective communication networks and technologically-driven response mechanisms.

Statement of Objectives

The study aimed to develop the IoT-Based Flood Alert System for Narvacan, Sto. Tomas, La Union.

Specifically, the study sought to achieve the following objectives:

1. develop the prototype of the IoT-Based Flood Alert System for Narvacan, Sto. Tomas, La Union; and
2. assess the system's level of acceptability using a user technology acceptability questionnaire.

Time and Place of Study

This study was conducted from August to April, in the school year 2023-2024, at Barangay Narvacan, Sto. Tomas, La Union, where the researchers conducted personal interviews with the residents and actual observations.

Definition of Terms

For a better understanding of this research report, the following terms were operationally defined:

Alert System is one of the features of the proposed system used to notify residents of the area of the risk of flooding. The alert notification is sent through SMS.

Alpha Testing was the phase of software testing carried out by internal teams, often the developers themselves and IT experts, to evaluate the quality, functionality, and overall performance of a software product before it was released to external users. This identified and addressed any issues or defects in the software to ensure a smooth and error-free experience for end-users.

Beta Testing was a phase of software testing conducted by real users, including external/end-users, in an unstructured environment. The software was released to a limited

audience outside the development team, allowing for real-world usage and feedback collection.

Central Server is used to store and analyze data from the sensors. The central server also generates alerts and sends them to residents of the area.

Coastal Risk Management is a strategy and process to assess, mitigate, and manage risks associated with coastal areas.

Communication Network is used to transmit data from the sensors to the central server. The communication network can be a wired or wireless network.

IOT stands for Internet of Things. The Internet of Things is a network of devices that are embedded with sensors, software, and network connectivity to collect and exchange data. In this case, the IoT-Based flood alert system likely used sensors to monitor water levels and other conditions that could indicate a potential flood. The data collected by these sensors would then be transmitted over the internet to a central location where it could be analyzed and used to issue flood alerts.

IOT-Based Flood Alert System was the proposed system that used the Internet of Things (IoT) to monitor and predict water levels. It consisted of a network of sensors that were deployed in areas at risk of flooding. The sensors collected data on water levels, rainfall, and other factors that could contribute to flooding. This data was then transmitted to a central server, where it was analyzed to generate flood alerts. The alerts could be sent to residents and authorities via SMS. This could give people time to evacuate or take other precautions to protect themselves and their property.

Monitoring Technologies were the advanced tools and systems used to monitor the

coastal environment, including potential hazards and changing conditions.

Real-time monitoring was a method for instantly assessing the current status of queues and channels within a queue manager, providing accurate information at the time of command execution. It involved querying real-time data using specific commands, facilitating prompt decision-making. Additionally, it extended to the collection of real-time data from physical entities, such as water levels, using prototypes or sensors, enabling proactive response strategies.

Sensors were used to collect data on water levels and rainfall. The sensors were installed in strategic locations throughout the area.

User Technology Acceptability Questionnaire (UTAQ) was a measure of how likely users were to adopt a new technology. UTA was influenced by a number of factors, including the perceived usefulness of the technology, the perceived ease of use of the technology, and the perceived compatibility of the technology with the user's existing values and beliefs.

Weather Forecast was a prediction of future atmospheric conditions. Weather forecasts were typically based on computer models that took into account a variety of factors, including current weather conditions, atmospheric pressure, temperature, humidity, and wind patterns. Weather forecasts could be issued for a variety of time periods, from a few hours to several days or even weeks in advance. One of the features that would be integrated in the web system.

Web-Based System was a software application that was designed to operate over the internet or an intranet. It was typically hosted on a remote server, and users accessed it

through a web browser using an internet connection. This was connected to the developed prototype as a means of displaying the status of water levels in the area and included a weather forecast.

Chapter 2

METHODOLOGY

Research Design

The researchers employed descriptive and developmental research designs in conducting the study.

A descriptive survey research design was used to describe the characteristics of a population, phenomenon, or situation (Chaudhari, 2021). This design aimed to collect data on the variables of interest and provide a detailed description. In the case of the IoT-Based Flood Alert System for Narvacan, Sto. Tomas, La Union, a descriptive survey research design was used to collect data based on the user's response to the system's level of acceptability using the User Technology Acceptability Questionnaire (UTAQ). Additionally, descriptive studies attempted to define, describe, or identify what was essential in developing the IoT-Based Flood Alert System for Narvacan, Sto. Tomas, La Union. This was accomplished by designing an interview guide questionnaire and survey that captured various variables of interest, such as water level rise, coastal erosion, and storm surge. The approaches for water level estimation and the use of sensors for collecting flood-related data were analyzed to provide functionality for the IoT-Based Flood Alert System.

In addition, Cruz and Dizon (2021) defined developmental research as a method that involved designing, developing, and evaluating instructional programs, processes, and products while adhering to specific standards and criteria. This approach held significant relevance in the realm of technology, especially for crafting prototype designs and

theoretical frameworks. They helped in analyzing and examining the design and helped to prepare to launch the idea. Mainly conducted by the designer, they made iterations and improved the quality of the design. They were usually progressive research based on the designs, making it easier for the designer to compile data and develop (Kumar, 2023). This study was crucial as it aimed to develop a prototype for the IoT-Based Flood Alert System for Narvacan, Sto. Tomas, La Union. Additionally, it sought to design and develop a web system specifically for the Municipal Disaster Risk Reduction and Management Office (MDRRMO). Correspondingly, the researchers used Agile methodology to serve as a guide in developing the said system.

Sources of Data

The study aimed to develop the prototype of an IoT-Based Flood Alert System for Narvacan, Sto. Tomas, La Union, enriched by a comprehensive range of data sources. Drawing from academic literature on IoT applications in flood monitoring and disaster management, as well as case studies in flood-prone regions, the research integrated established best practices and technological advancements into the system's design. Primary data were gathered through field surveys and interview questionnaires conducted with local authorities, including the barangay official councils and the head of MDRRMO, as well as residents from each of the seven sectors of Barangay Narvacan, and stakeholders. These interactions provided valuable insights into the region's topography, historical flood patterns, and community-specific challenges. This contextual data served as a foundational element, enabling the construction of a flood alert system that not only incorporates

cutting-edge technology but is also tailored to the unique characteristics of Narvacan, Sto. Tomas, La Union. Subsequently, the system's level of acceptability was assessed using a user technology acceptability questionnaire while the performance of the water level sensor implemented through arduino technology was evaluated ensuring its practical suitability for the community. Purposive sampling was employed to select the sample population. The researchers administered the questionnaire to a sample population of thirty (30) respondents, including nine (9) barangay council officials and twenty-one (21) residents from various sectors in Barangay Narvacan, Sto. Tomas, La Union, as reflected in Table 1.

Table 1. Distribution of Respondents

Respondents	N
Barangay Official Councils	9
Sector 1	3
Sector 2	3
Sector 3	3
Sector 4	3
Sector 5	3
Sector 6	3
Sector 7	3
Total	30

Materials and Procedures

The researchers used a variety of materials and procedures to achieve the objectives of the study.

Researchers utilized the Agile Methodology (see Figure 1 on page 27) as their guiding framework in developing the prototype of the IoT-Based Flood Alert System for Narvacan, Sto. Tomas, La Union.

Agile software development comprised a set of values, principles, and practices that could be effectively applied to software development projects. This methodology strongly emphasized process adaptability and customer satisfaction by rapidly delivering functional software products. Additionally, it promoted collaborative decision-making, ensuring customer satisfaction throughout the development process of the IoT-Based Flood Alert System. Agile had gained popularity due to its flexibility and adaptability, streamlining software development by eliminating unnecessary barriers and maximizing flexibility, quality, and user involvement. It facilitated the performance at the speed of relevance (Harris, 2020).

This approach ensured a comprehensive and user-centric system creation process by incorporating well-defined phases such as requirements, design, development, testing, deployment, and review. This iterative approach ensured continuous user feedback and system improvement, resulting in a more effective and user-centric flood alert system. Additionally, Agile methodology emphasis on collaboration and flexibility enabled the team to swiftly address and resolve issues, ensuring a high-quality and reliable system.

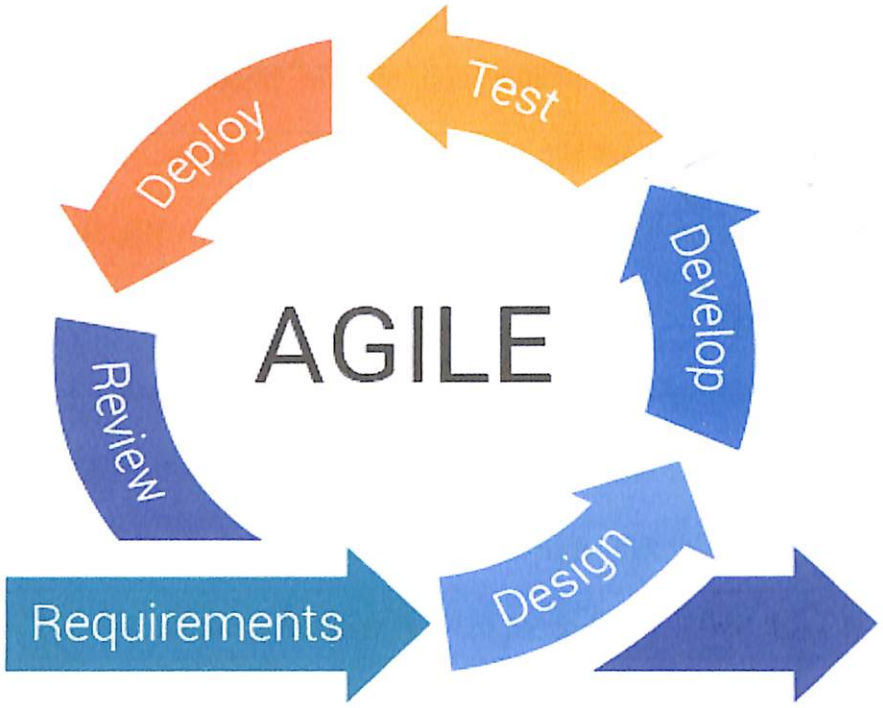

Fig. 1. Agile Methodology (Harris, 2020)

Requirements. The first phase was dedicated to requirement gathering and analysis. During this phase, the researchers conducted personal interviews with the head of MDRRMO and the captain of Barangay Narvacan, Sto. Tomas, La Union. Through this, the researchers determined the tools they used when floods came, as well as the present problems they encountered. These requirements served as features to be included in developing the IoT-Based Flood Alert System.

Design. In this phase, the researchers meticulously crafted a detailed plan for the structure, prototype, interface, and strategies. Additionally, they thoroughly examined the system's features and prototype. This phase defined the roadmap for development, ensuring that the system met the intended objectives and user requirements. It served as a bridge between the planning and development phases, including the implementation of use case diagrams and flowcharts, setting the stage for the actual coding and implementation of the system.

Develop. In this phase, all the components of the designed software were implemented, and the source code was created. After the required information had been collected, the focus shifted from planning and design to the actual implementation of the system prototype. In the construction of the IoT-Based Flood Alert System, the researchers employed various software and hardware tools. Arduino Uno r3 Drip was utilized as a hardware tool connected to ultrasonic sensor to analyze the water level and a temperature sensor in predicting the water depth level the Circuit Rocks Breadboard served as a spark for hobbyists. Then, these analyzed values were passed to the Arduino Uno. Additionally, several software tools and frameworks were maximized, including C/C++, HTML, CSS, PHP, MySQL, and JavaScript, for an intuitive user interface and efficient data management. Consequently, the Arduino passed the alert message to the IoT module. These analyses showed how the prototype was embedded in real-time for prediction and sending alerts.

Test. This phase involved checking any faulty parts of the code and their fixes. It focused on evaluating the system's usefulness, information quality, and interface quality. Various testing activities ensured that the product met the desired quality standards and aligned with user requirements. The developers themselves performed alpha testing by using the system as if it were being used in a work environment. After successful alpha testing, the software underwent beta testing, where it was released to a selected group of end-users/IT experts to be used in their production environment. Beta testing aimed to thoroughly assess the software and hardware product's quality, functionality, and performance under real-world conditions. Using the User Technology Acceptability

Questionnaire (UTAQ), the IoT-Based Flood Alert System was thoroughly examined, and any remaining defects were identified. Verification was conducted to ensure that the product met the specified requirements and delivered value to the intended users.

Deploy. After complete testing, the IoT-Based Flood Alert System was ready for implementation in this phase. The goal was to ensure the system implemented real-time water level monitoring and continuously tracked the high-water level risk. The primary objective was establishing the system's functionality within the target location. This involved setting up the system for real-time water level monitoring. Sensors were meticulously installed in strategic locations to ensure comprehensive area coverage. These sensors continuously collected data, which were analyzed to track water levels and identify potential flood risks in Barangay Narvacan, Sto. Tomas, La Union.

Review. After deployment and system implementation, there were circumstances where errors occurred. This phase involved close monitoring of the system's performance. Data quality, functionality, and any error messages were continuously reviewed. If errors were identified, the developers stepped in to diagnose and fix the problems. This might have involved troubleshooting hardware or software issues. The developers resolved any problems. In the ongoing development and operations of the system, continuous support was provided to ensure it operated effortlessly. The researchers established clear communication channels with the stakeholders in Barangay Narvacan, Sto. Tomas, La Union, including the local government and residents. This allowed users to report any issues they encountered or suggest improvements. The developers then analyzed this feedback and adjusted to optimize the system's performance and user experience.

For the second objective, the researchers used the User Technology Acceptability Questionnaire (UTAQ) through purposive sampling to assess system's level of acceptability and user satisfaction. This method involved carefully selecting participants with direct experience or knowledge in flood monitoring, disaster management, or the implementation of IoT-based systems in the target area. This ensured valuable feedback from users who understood the context and could effectively assess the system's effectiveness. Purposive sampling was a non-probability method for obtaining a sample where researchers used their expertise to choose specific participants that helped the study meet its goals. These subjects had particular characteristics that the researchers needed to evaluate their research question. In other words, the researchers picked the participants "on purpose" (Frost, 2023).

Data Gathered

Throughout the development life cycle, the researchers gathered the following information needed to achieve their objectives in the study. In the requirement phase, the data gathered included the intended purpose of the prototype, features, and user requirements specification, as well as the material requirements to develop the prototype. In the design phase, data gathered comprised user requirements, user interface design, and user experience design. The data gathered in the develop phase encompassed coding, debugging, and testing in the various tools and applications. In the test phase, data gathered included the outcomes of functional testing, performance testing, and acceptance testing.

Whereas, in the deploy phase, data gathered included user feedback for ongoing refinement and optimization of the prototype.

Lastly, after the deployment of the IoT-Based Flood Alert System, the users evaluated the technology acceptability in terms of a) system usefulness, b) information quality, and c) interface quality. This was utilized as an input to assess the level of acceptability of the prototype that had been developed.

Data Analysis

To evaluate and analyze the data gathered from respondents, it was tallied, classified, and analyzed. Results from the survey questionnaire were computed using frequency counts and average weighted mean, and interpreted using the seven-point scale in the User Technology Acceptability Questionnaire.

The AWM formula is:

$$AWM = \frac{\sum WM}{N}$$

$\sum$ = denotes the summation

WM = is the weighted mean

N = is the total respondents

As posited by Vaidya (2024), the weighted mean equation was a statistical method used to calculate the average by multiplying the weights with their respective means and taking their sum. It was a type of average in which weights assigned individual values to

determine the relative importance of each. Respondents were asked to rate their agreement with the statements, ranging from strongly disagree to strongly agree, using the Likert Scale for the user technology acceptance of the prototype.

Scale	Statistical Range	Descriptive Equivalent Rating	Descriptive Interpretation
7	6.16 – 7.00	Strongly Agree	Extremely High Acceptance
6	5.30 – 6.15	Agree	Very High Acceptance
5	4.44 – 5.29	Somewhat Agree	Moderately High Acceptance
4	3.58 – 4.43	Neutral	Neutral Acceptance
3	2.72 – 3.57	Somewhat Disagree	Moderately Low Acceptance
2	1.86 – 2.71	Disagree	Low Acceptance
1	1.00 – 1.85	Strongly Disagree	Extremely Low Acceptance

Mean responses were interpreted depending on each characteristic. Responses with a mean ranging from 1.00 – 1.85 were interpreted as Extremely Low Acceptance, indicating that the IoT-Based flood alert system did not conform to its intended functions and objectives. Moreover, responses with a mean rating ranging from 1.86 – 2.71 were interpreted as Low Acceptance, denoting that the respondents were quite satisfied with the features offered by the system. The interpretation was Moderately Low Acceptance if the responses had a mean rating of 2.72 – 3.57, indicating that the respondents may have had concerns about the technology's usefulness, ease of use, compatibility, or affordability. While the mean rating ranging from 3.58 – 4.43 was interpreted as Neutral Acceptance, suggesting that respondents thought the system fulfilled its features. The researchers

interpreted responses with a mean rating range of 4.44 – 5.29 as Moderately High Acceptance, meaning the respondents agreed that the system performed well and met the indicated criteria in each indicator. While the mean rating ranging from 5.30 – 6.15 was interpreted as Very High Acceptance, indicating that users found the technology useful, easy to use, compatible with their values, and affordable. Lastly, the responses were interpreted as Extremely High Acceptance if the mean rating fell between 6.16 – 7.00, signifying that the respondents strongly agreed that the proposed prototype performed all the required features and functionalities in which it achieved satisfactorily all the different criteria.

Ethical Consideration

The researchers applied the principles of ethical considerations following the framework developed by Bhandari (2021). The framework underscored the critical importance of upholding a precise code of ethics in gathering data from individuals. This commitment ensured the protection and rights of all research participants, bolstered the credibility of the research, and upheld the integrity of the scientific and academic process.

In this study's context, utmost care was taken to avoid any harm to participants. Before commencing the study, comprehensive consent was sought from all participants of Barangay Narvacan, Sto. Tomas, La Union. Additionally, preserving the confidentiality and security of personal information and the identities of all research participants was a top priority. Their privacy was rigorously safeguarded throughout the study, and all data collected was handled with the highest level of discretion.

Throughout the entire research process, ethical considerations remained of the utmost importance. The researchers prioritized participant safety and well-being by obtaining approval from the DMMMSU-Research Ethics Committee (as described in Appendix H) and receiving informed consent forms from all participants (as described in Appendix J and K). These consent forms provided comprehensive information about the study, including its goals, methods, possible advantages and disadvantages, and participant rights.

The principle of voluntary participation served as the bedrock of this research effort. Each respondent involved in the study possessed the inherent right to engage willingly and retained the freedom to withdraw from the study at any stage, should they choose to do so. This unwavering emphasis on voluntariness reaffirmed the researchers' dedication to conducting research that was not only methodologically robust but also deeply respectful of the autonomy and choices of those who contributed to its success.

Chapter 3

RESULTS AND DISCUSSION

Development of an IOT-Based Flood Alert System

The Arduino was deemed appropriate for the system referencing the work of Muaz (2021), who conducted research on sampling and analyzing water river flow conditions and characteristics using the agile methodology and applying the same model.

The researchers constructed the prototype using a combination of hardware components and electronic modules. At its core, the Arduino Uno r3 Drip Type served as the central processing unit, governing various functions based on input from sensors and modules. Facilitating wireless connectivity, the ESP-01s WiFi module enabled the system to communicate with other devices or access the internet. Male to Male Jumper Wires and Male to Female Jumper wires were employed to establish electrical connections between different components on the breadboard and between components to the Arduino board. The Ultrasonic Ranging Sensor utilized ultrasonic waves to accurately measure distances between the sensor and objects. These sensors, likely ESP-01 microcontrollers, transmitted data to the Blynk cloud platform which acted as an intermediary for processing and storage before forwarding it to the designated recipient or application for further analysis and action. A Circuit rocks Breadboard 400 provided a platform for prototyping and connecting electronic components without soldering. Ultimately, a 12V Active Buzzer served as an output device, producing audible sound signals to provide feedback or alerts based on detected conditions or events. Together, these components formed the foundation of the

system, enabling it to sense its environment, wirelessly communicate, process data, and provide feedback through sound signals. The researchers adopted the following phases in the development of the system.

Requirements. In the initial phase of the research, the researchers engaged in personal interviews following unstructured questions (see Appendix A) with key stakeholders, including the captain of Barangay Narvacan, Sto. Tomas, La Union, as well as all the barangay officials and three residents in each of the seven sectors of the barangay. These interviews were crucial for understanding the challenges associated with the manual processes of flood and high tide monitoring. The researchers prioritized ethical considerations in selecting their participants, ensuring that those involved were genuinely impacted by the issues at hand and had a vested interest in the proposed solution.

Through these interviews, it became apparent that the current approach to monitoring water levels relied heavily on manual and traditional methods. The researchers maintained constant communication with the stakeholders, allowing them to delve deeply into the intricacies of the existing monitoring practices. One significant revelation was that high tide and low tide occurrences were manually predicted, an approach susceptible to human error and limited by the availability of resources. This discovery underscored the urgency of modernizing the monitoring process to enhance accuracy, efficiency, and timeliness in responding to flood and high tide events.

Furthermore, through these interactions, the researchers gleaned insights into the specific functionalities required of the developed system. Understanding the stakeholders' needs and challenges guided the development process, ensuring that the system would

address critical pain points effectively. The researchers identified the need for complementary tools and prototypes to augment the functionality of the monitoring system, thereby enhancing its overall effectiveness in mitigating the impact of natural disasters.

By integrating ethical considerations into their approach and actively involving stakeholders in the process, the researchers laid a solid foundation for the development of a tailored solution to the challenges posed by manual flood and high tide monitoring. This collaborative effort not only facilitated a deeper understanding of the issues at hand but also fostered a sense of ownership and investment among the stakeholders, paving the way for meaningful and sustainable change.

Design. Upon gathering the necessary information, the researchers embarked on the task of translating their insights into tangible system designs. They initiated this process by drawing use case diagrams and flowcharts, crucial tools for illustrating the organizational structure of the system. With a detailed plan for the structure, prototype, interface, and strategies, the researchers used iterative cycles of designing, prototyping, and refining the system components, ensuring continuous feedback and improvements to meet the project requirements.

Parallel with the design phase, the researchers began developing prototypes to validate their conceptual designs and demonstrate key functionalities. These prototypes served as tangible representations of the envisioned system, allowing stakeholders to interact with and provide feedback on its usability and effectiveness. Utilizing materials such as wireframes, mockups, and simulation software, the researchers iteratively refined the prototypes based on user input, ensuring that the final product would meet the

requirements and expectations of its intended users. Moreover, these diagrams served as invaluable conversation starters during engagements with stakeholders. They facilitated discussions around system requirements, eliciting feedback that was instrumental in refining the design and enhancing the user experience.

Using user views as their primary focus, the researchers created a use case diagram as a visual aid for outlining the functional needs of the system. This figure depicted the dynamic interactions between users (actors) and the system, presenting several use cases or situations in which users interacted with the system to achieve particular goals. Clear communication and understanding among stakeholders were facilitated by the use case diagram, which provided interactions and a thorough overview of the system's functionality and user interactions.

As illustrated in Figure 2, the diagram displays the various use case scenarios of flood alert system designed to keep residents and barangay official councils informed about potential flooding. It includes actors namely residents/users, barangay official councils and the head of MDRRMO. Residents and barangay official councils could access the system through a website that retrieved real-time water level data from sensors deployed near waterways, receive SMS notifications and monitor the water level. The head of MDRRMO (Municipal Disaster Risk Reduction and Management Office) can log in to the website to view recorded water levels, generate reports, access the website manual, utilize printing functions, and log out of the web system. Additionally, monitoring of water levels is being integrated to collect real-time data. The website could then visualize historical water level information, potentially presented as monthly reports, for residents and barangay officials.

When water levels reached a critical threshold, the system generated alerts. In essence, this system aimed to provide residents and barangay official councils with timely information and critical flood warnings through a web-based interface.

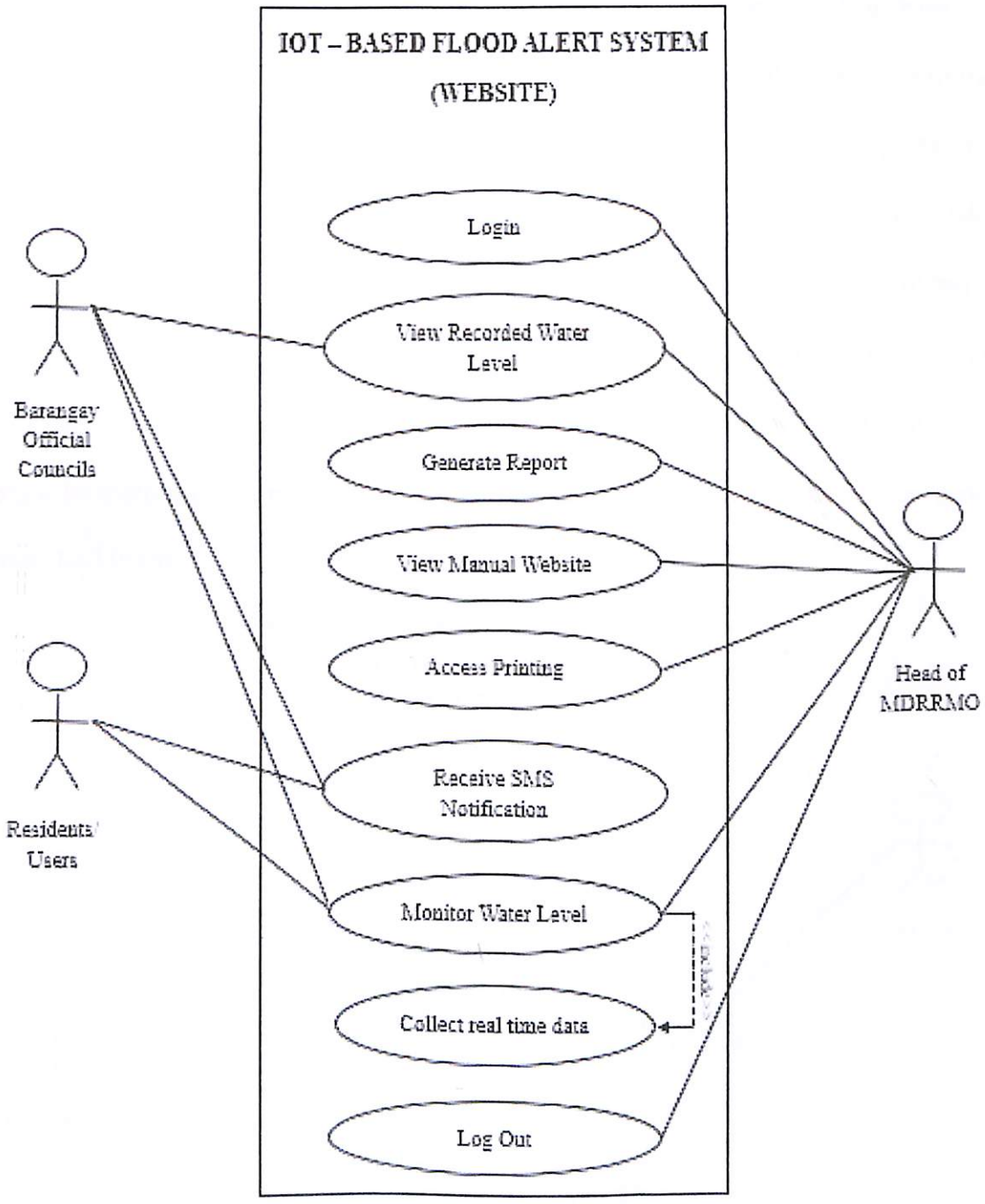

Fig. 2. Use Case Diagram (Website)

The diagram in Figure 3 illustrates the flow or process how the barangay official councils, residents/users and arduino as in formed about impending flooding. Residents and barangay official councils might receive SMS message and access real-time water level data collected by strategically placed sensors near water bodies using an easy-to-use online interface. Arduino send SMS message and acting as a conduit between the sensors and the web interface, these sensors most likely transmitted data to Blynk Cloud. Most importantly, the system alerted locals when water levels rose to dangerous levels, enabling them to take precautions against impending flooding. Although the precise purpose of the third-party programs was not specified, it is possible that the website integrated with them. The overall goal of this system was to employ an intuitive web interface to deliver critical flood alerts and timely information to barangay official councils and residents of Barangay Narvacan, Sto. Tomas, La Union.

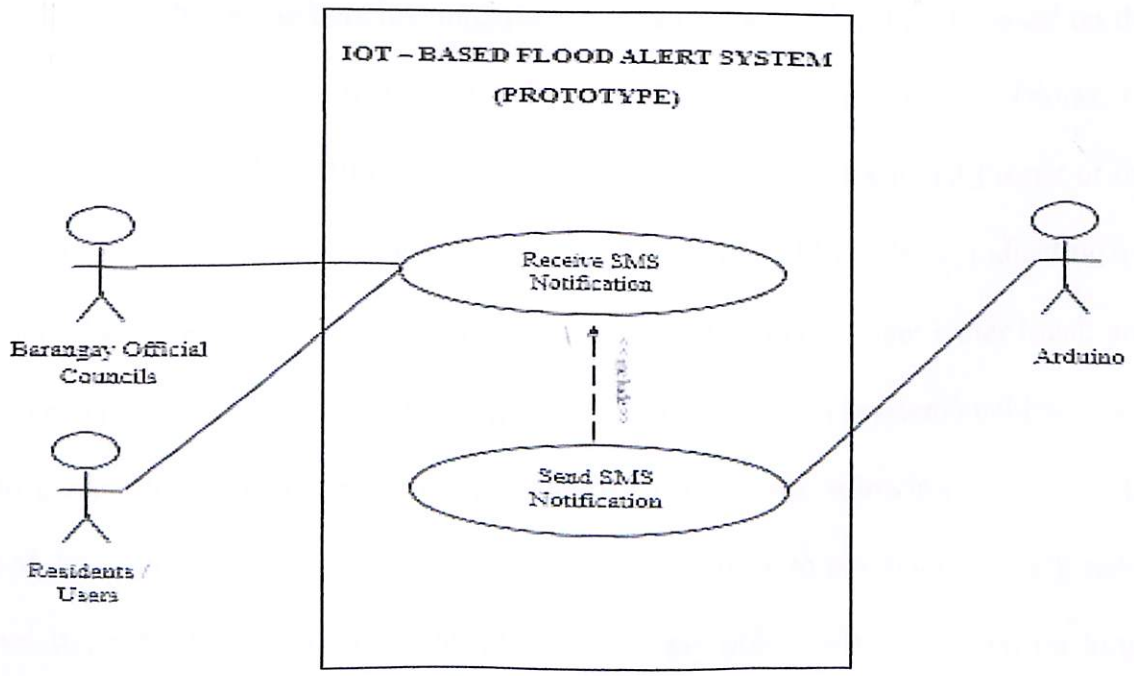

Fig. 3. Use Case Diagram (Prototype)

The diagram in Figure 4 showcases a more complex flood alert system, designed to keep residents and local government informed about potential flooding threats, compared to a previously proposed system. This system utilized various components to collect, process, and deliver water level data. Sensors, likely ultrasonic (HC-SR04), deployed near waterways measured water levels and transmitted data to an Arduino Uno board. This board might have performed some initial processing before potentially using an ESP8266 chip for wireless transmission. The Blynk cloud platform acted as a central hub, potentially receiving processed data from the ESP8266 chip over Wi-Fi. This data was stored for later access and analysis. While the specific role of Node-Red, an open-source programming tool, was involved in data processing and web interface visualization, the system generated critical flood alerts when water levels reached pre-defined thresholds of 1 to 3.

However, the researchers had integrated a localized water level scale based on the observation and measurement of the water level in Barangay Narvacan, Sto. Tomas, La Union, while the SMS text notifications were being adapted to the localized format of the MDRRMO, consisting of three levels: Alert Level 1 represented by yellow, indicating a 1 feet water level; Alert Level 2 represented by orange, indicating a 2 feet water level; and Alert Level 3 represented by red, indicating a 3 feet water level. This system enables timely alerts to be sent to residents based on the observed water levels, allowing them to make informed decisions and take necessary precautions in response to potential flooding risks. Additionally, a database system like MySQL could have potentially been used for long-term data storage and analysis. In essence, this intricate system aimed to collect real-time

water level data, transmit it wirelessly, store it in the cloud, potentially analyze trends, and generate critical flood alerts. By providing timely information, this system could have been a valuable tool for flood preparedness in the community.

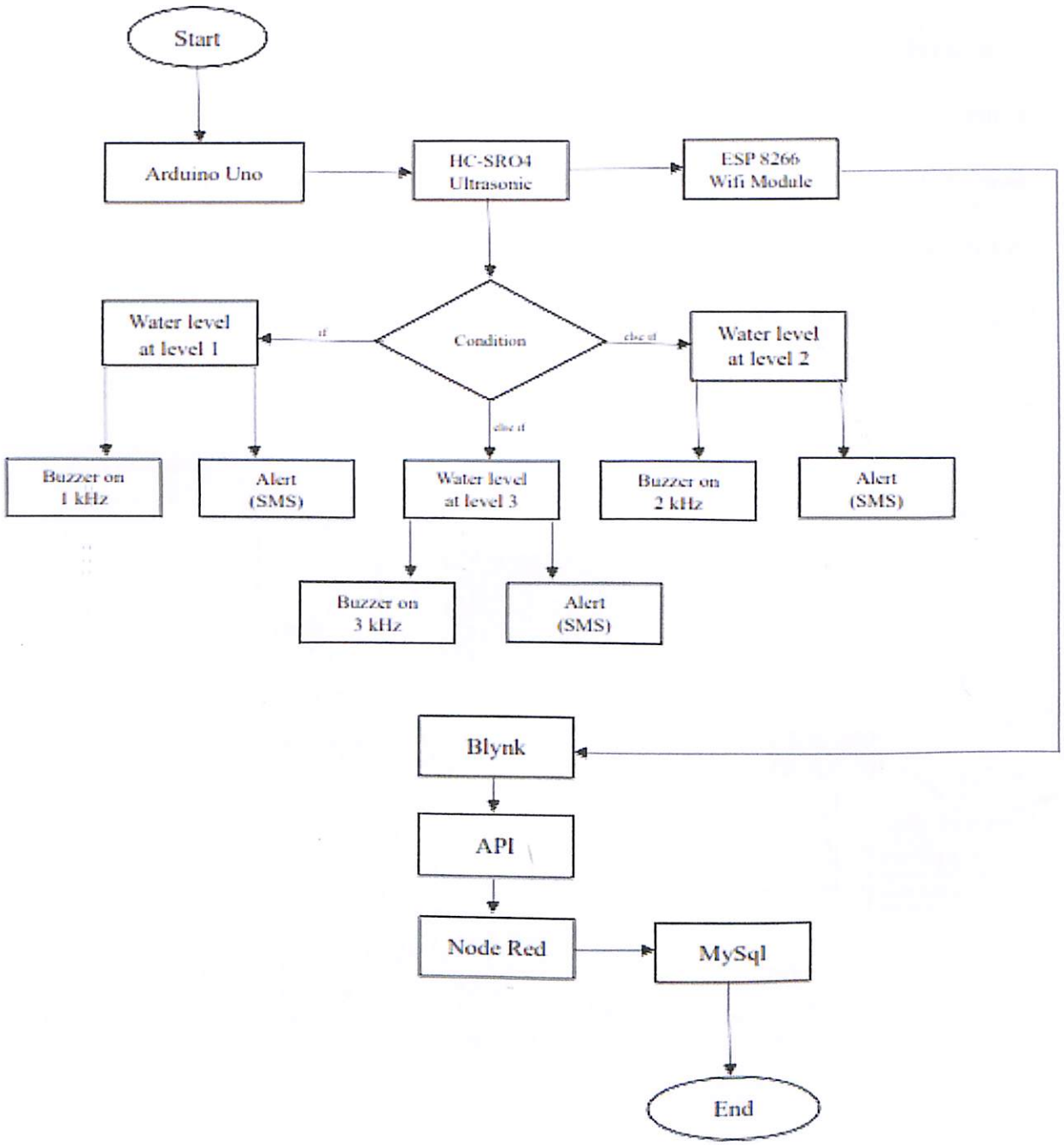

Fig. 4. Flowchart of the Developed Prototype

The Schematic Diagram of Flood Alert System Prototype in Figure 5 comprises an Arduino Uno as the control center, interpreting data from a downward-positioned ultrasonic sensor via jumper wires on a breadboard. Upon water detection, the Arduino triggers a buzzer alarm. Integration of an ESP-01S WiFi module allows for remote notifications or cloud-based data logging through WiFi connection. Additionally, incorporating a battery for power supply ensures system reliability and autonomy, crucial for continuous monitoring in flood-prone areas. While the schematic outlines basic functionality, specific connections, and programming may vary depending on the chosen guide or tutorial for building this prototype, allowing for customization based on individual needs and preferences.

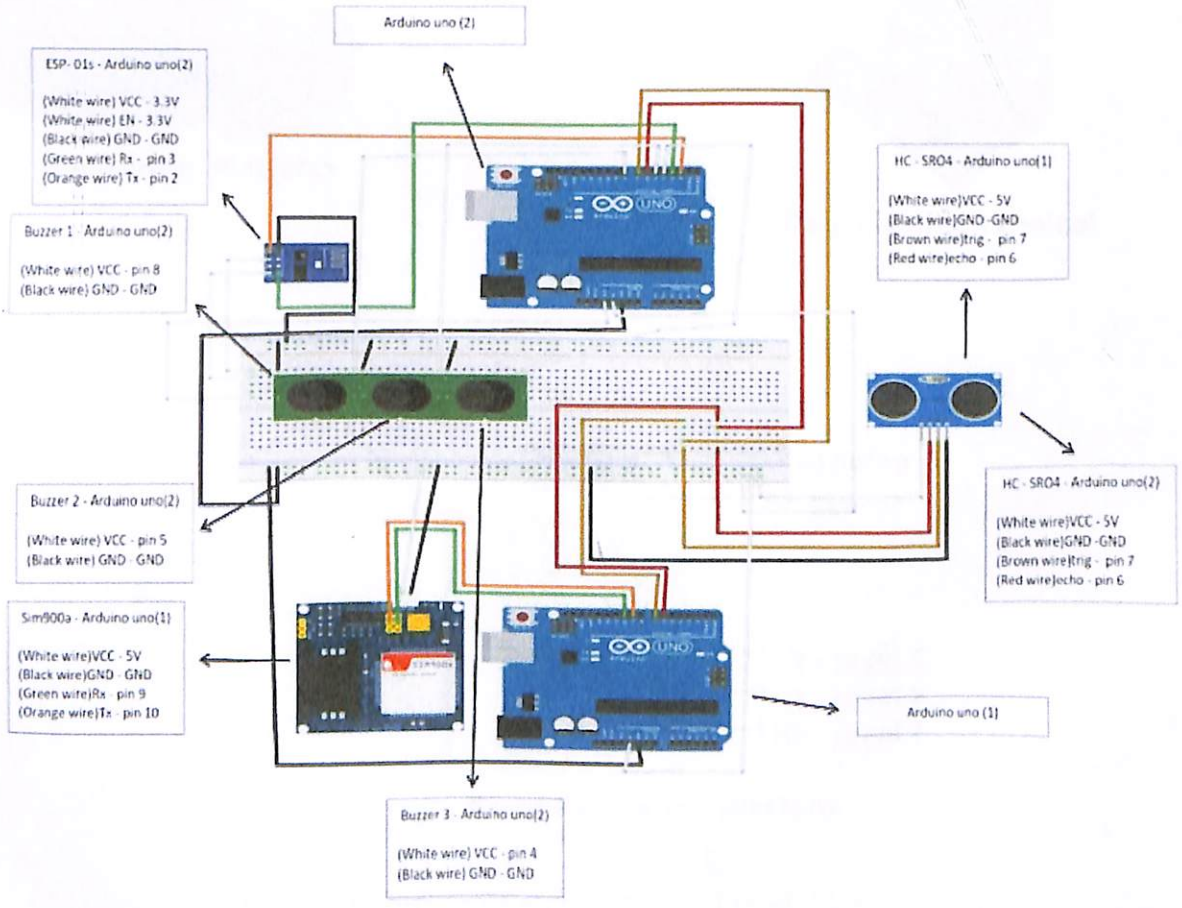

Fig. 5. Schematic Diagram Structure of an IoT-Based Flood Alert System

Figure 6 shows an IoT-based flood alert system prototype. In this setup, an ultrasonic sensor is responsible for detecting water levels, transmitting data to the Arduino Uno microcontroller. Upon detecting water, the Arduino triggers a buzzer to sound an alarm and relayed the data to a monitoring platform accessible by the MDRRMO, which is likely a disaster response agency, for viewing. This integration of hardware components with data transmission capabilities exemplifies the versatility and potential applications of IoT systems in addressing environmental challenges and enhancing disaster preparedness and response efforts.

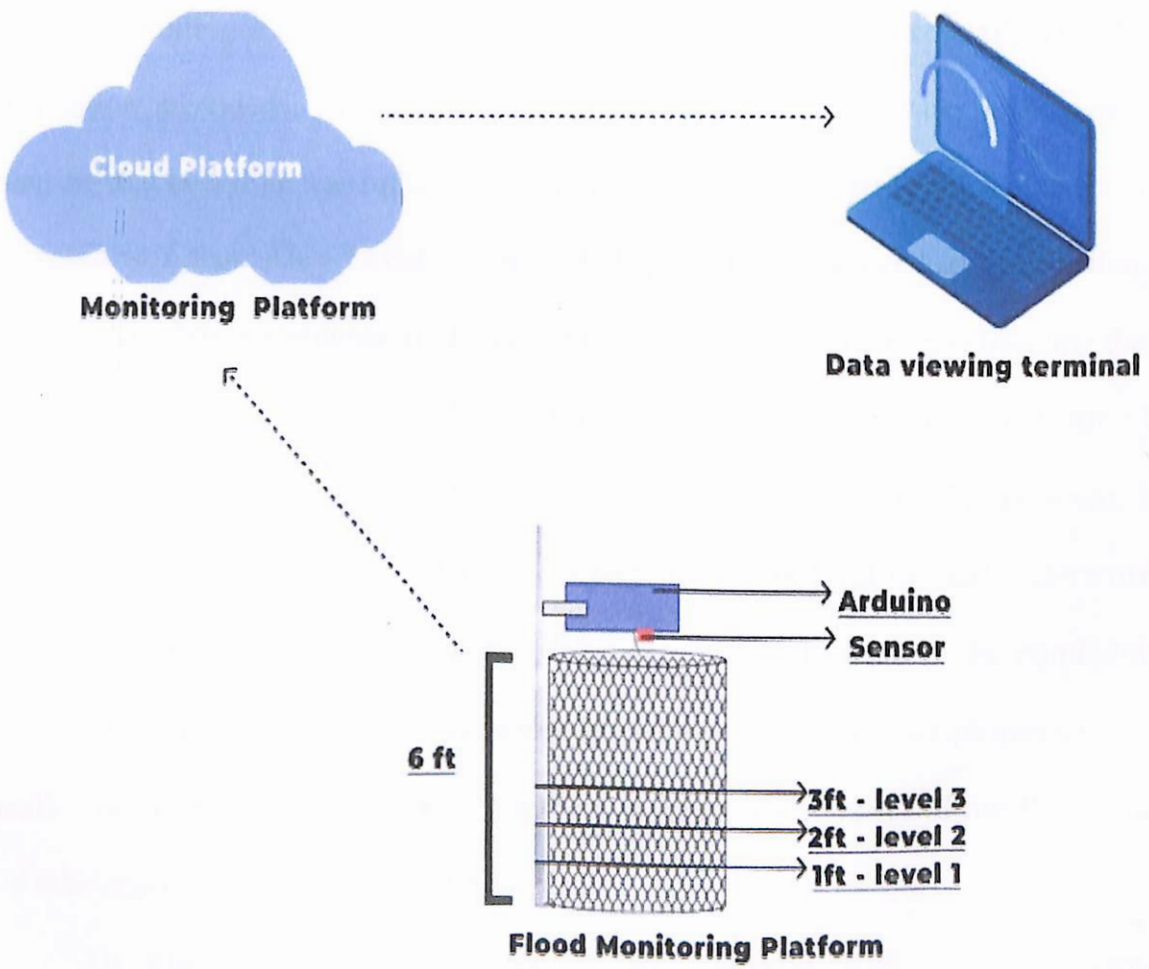

Fig. 6. Prototype of an IoT-Based Flood Alert System

In Figure 7, the external interface of the IoT-Based Flood Alert System reveals the utilization of key materials including a Pole, Mesh Net, and Handle. Notably, the customized housing of an Arduino was prominently positioned atop the pole, approximately 6 feet from the ground. This strategic placement ensured optimal data collection from the environment while safeguarding the device from potential water damage.

Additionally, the figure illustrates the incorporation of three distinct alert levels, as shows in Figure 4 on page 42. When the water level rises to 1-foot, Alert Level 1 is triggered, sending a text notification to residents and barangay council officials. This notification provided a warning along with the corresponding water level percentage, ensuring that everyone was informed about the initial rise in water level. As the water level increases to 2 feet, Alert Level 2 is activated, prompting another text notification. This notification advises residents to prepare to move to higher ground, emphasizing the need for heightened awareness and readiness. When the water level reached 3 feet, Alert Level 3 mandated an evacuation, directing residents, especially those in affected areas, to the nearest evacuation center (see Plate 17 on page 66). This level of alert underscores the seriousness of the situation, prioritizing the safety and well-being of the community by ensuring timely and organized evacuation procedures. These alert levels serve to provide detailed notifications to users, enabling them to gauge the severity of the flood situation and take appropriate action accordingly.

To effectively implement this system, several strategies are employed. Collaboration with the Barangay Official Councils of Narvacan, Sto. Tomas, La Union and

the Municipal Disaster Risk Reduction and Management Office (MDRRMO) of Santo Tomas ensured that the system aligned with existing emergency protocols. Community engagement programs were conducted to educate residents about the system's benefits and usage. Data redundancy and backup mechanisms were established to maintain system reliability during power outages or network failures. Engaging with community leaders, such as barangay captains and councils, helped ensure effective dissemination of information and encouraged community participation. Regular maintenance and calibration of sensors were planned to ensure accuracy and longevity. Additionally, the system was continuously evaluated and improved based on feedback from stakeholders, including local authorities, disaster management teams, and community members. This iterative process led to enhancements in system functionality, user experience, and overall effectiveness in mitigating flood-related risks.

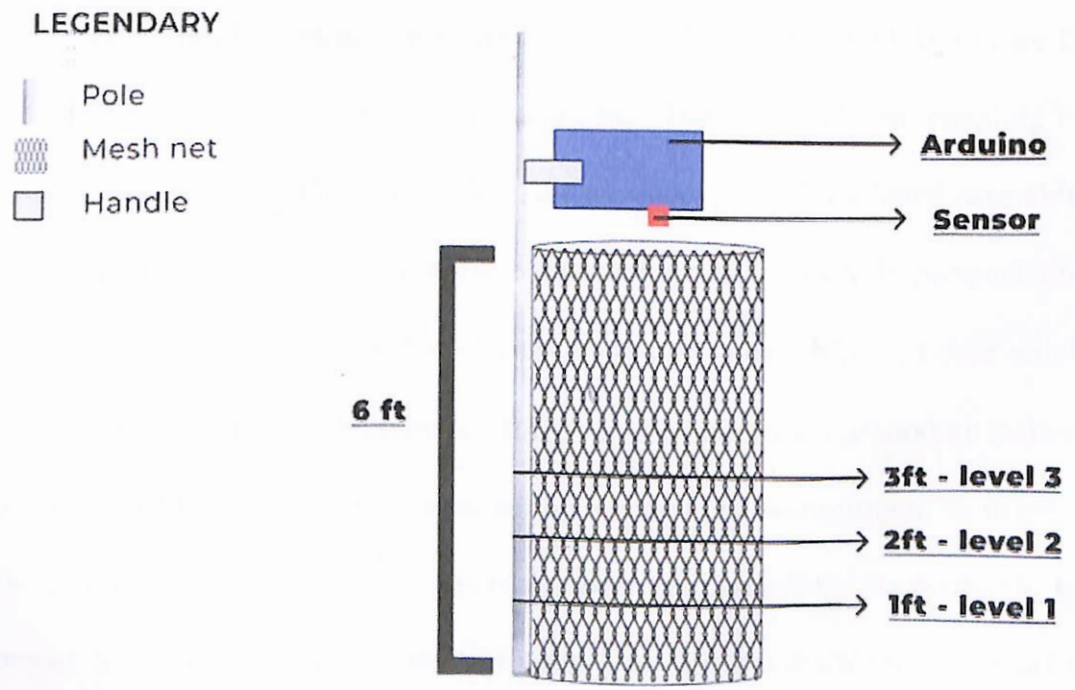

Fig. 7. External Interface of an IoT-Based Flood Alert System

Develop. The researchers implemented the functions and features that were needed by the client. They utilized an Arduino Uno to establish a connection for monitoring water flow alongside sensors to analyze water parameters. Furthermore, they simulated Arduino behavior, conducted real-time data collection, and integrated these data streams into a developed website. In the develop phase of the IoT-Based Flood Alert System, the researchers utilized various hardware and software tools. This website, crafted with C/C++, HTML, CSS, PHP, MySQL, and JavaScript, not only facilitated an intuitive user interface but also ensured efficient data management. As part of this integration, analyzed values were transmitted to the Arduino, which played a pivotal role in passing alert messages to the IoT module.

Moreover, the researchers demonstrated the practicality and reliability of their solution by incorporating functionalities such as text message alerts. They extended the system's reach by implementing a website for the Head of MDRRMO, Barangay Official Councils, and residents of Barangay Narvacan, Sto. Tomas, La Union, enabling them to monitor live water levels. This comprehensive approach, which included assembling the Arduino, simulating its behavior, real-time data collection, website implementation, and targeted user access, underscored the effectiveness and applicability of their solution in addressing water monitoring challenges. Moreover, the integration of modern technologies with practical field deployment showcased the researchers' commitment to developing a holistic and user-friendly solution. Furthermore, the website's accessibility enabled widespread dissemination of critical information, fostering a collaborative effort toward proactive flood management and community preparedness.

Table 2. Hardware and Software Components Used in Prototype Development

Hardware	Description
Arduino Uno r3 Drip Type	is used to monitor water levels and trigger warnings when potential flooding is detected. It collects data from water level sensors and activates alerts like buzzers, lights, or notifications based on predefined thresholds to warn residents and authorities.
Circuit Rocks Breadboard 400	is used for prototyping and testing electronic circuits. It allows researcher to assemble and connect components like ultrasonic ranging sensors, Arduino Uno R3 microcontrollers, and alert mechanisms.
Ultrasonic Ranging Sensor	it measures water levels in flood-prone areas by emitting ultrasonic waves and calculating the time. It provides real-time data for flood alert systems to issue timely warnings and preventive actions.
ESP-01S WiFi Module	are wireless communication for flood alert systems, allowing them to send real-time alerts and data to remote servers and devices during flood events.
Active Buzzer	is a flood alert system emitting audible alarms when water levels surpass predefined thresholds, signaling residents and authorities for immediate action.
Male to Male Jumper Wire	establish connections between electronic components, enabling data transmission and signal communication for monitoring water levels and triggering alerts during flood events.
Male to Female Jumper Wire	connect male pins on sensors or modules to female ports, enabling seamless integration and communication between components.
GSM Module SIM900A	transmits real-time water level data from sensors to a central server and sends SMS alerts to authorities and residents when flood thresholds are reached.

Continuation of Table 2

Hardware	Description
18650 Battery 3.7v li-ion	is a high energy density and long cycle life, are ideal for IoT-based flood alert systems, ensuring reliable, long-lasting power.
Software	**Description**
Blynk Cloud (API)	is used to remote monitoring and control of a flood alert system through the Blynk mobile app or web dashboard, allowing users to access real-time data and trigger actions from anywhere with internet access.
Node GS (Red)	is a ground station in a flood alert system, receiving and processing data from remote sensors to monitor water levels and terrain data, enhancing flood prediction and response capabilities.
Arduino IDE	is used in a flood alert system to program Arduino microcontrollers, enabling control of sensors, actuators, and communication modules essential for system operation.

The components listed were essential elements of a flood alert system designed to monitor water levels, transmit data, and issue timely warnings to residents and barangay official councils of Barangay Narvacan, Sto. Tomas, La Union. The Arduino Uno R3 controlled the system, collecting sensor data and triggering alerts based on thresholds. The Circuit Rocks Breadboard and jumper wires enabled prototyping and interconnecting components. The 18650 Battery supplied power backup. Blynk Cloud API, Node GS, and Arduino IDE enabled remote monitoring, data processing, and programming for efficient flood prediction and response.

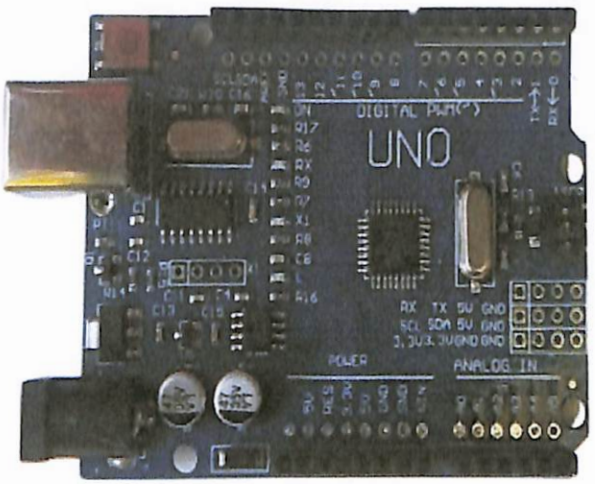

Plate 1. Arduino Uno r3 Drip Type

Plate 1 shows the arduino uno rev3 at the core of an obstacle detection system. It received data from an ultrasonic sensor, interpreting distance using sound waves and reacting to object proximity. Through its digital and analog pins, the Uno connected to the sensor, allowing for alarms or other actions to be triggered when an object came too close. The Arduino Uno Rev3's widespread adoption stemmed from its robust feature set and user-friendly design, making it accessible to both beginners and experienced developers. Its open-source platform fostered a vibrant community contributing to a vast repository of code libraries, tutorials, and project ideas. The Arduino's ability to handle digital and analog inputs and outputs made it particularly suited for managing a wide range of sensors and output devices. Its compatibility with various programming languages and integration with other hardware components further enhanced its capabilities, allowing for seamless expansion and customization in countless applications.

Plate 2. Male to Female Jumper Wires

Plate 2 presents the male to female jumper wires, which were likely utilized to connect various components, forming the backbone of the system's interconnectivity. These versatile wires, typically featuring a male pin on one end and a female connector on the other, facilitated straightforward connections between the Arduino Uno's pins, the ultrasonic sensor, and the battery pack. This design ensured secure and reliable connections for both data transmission and power flow within the obstacle detection system. The use of jumper wires offered several advantages, particularly in prototyping and development, by providing flexibility and reusability, saving time, and allowing for rapid iteration without the need for soldering. The color-coded insulation of jumper wires also helped in identifying and organizing connections, reducing wiring errors and enhancing the overall clarity of the circuit layout.

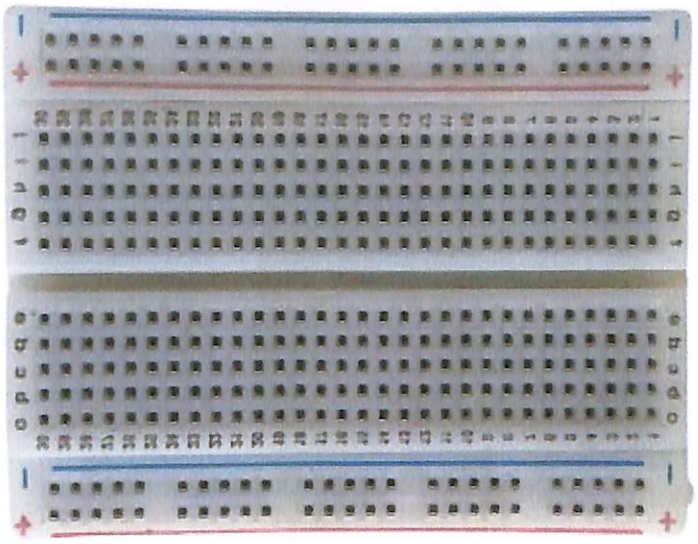

Plate 3. Circuit Rocks Breadboard 400

Plate 3 showcases the circuit rocks breadboard 400, which was more than just a standard grid of holes; it served as a spark for hobbyists and electronics enthusiasts to express their creativity. This multipurpose tool revolutionized experimentation and prototyping by doing away with the necessity for soldering. One could just picture putting integrated circuits (ICs), resistors, and capacitors on the breadboard and connecting them with jumper wires. This flexibility made it possible to quickly construct and test various circuit configurations. It gave users the ability to quickly troubleshoot problems and refine their designs before committing to a permanently soldered board. The breadboard's modular nature allowed for a level of experimentation that was previously unattainable, opening up new avenues for innovation and learning. Its user-friendly design encouraged beginners to dive into the world of electronics without feeling overwhelmed, while seasoned professionals appreciated the convenience and efficiency it offered.

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Plate 4. Male to Male Jumper Wires

Plate 4 illustrates the male-to-male jumper wires, which played a crucial role in connecting various components on the breadboard, featuring male connectors on both ends that made it convenient to establish connections between sensors, the Arduino board, and other modules. In an Arduino-based obstacle detection system, these wires facilitated the connection of the ultrasonic sensor to the Arduino Uno, enabling the sensor to receive power and send data signals back to the microcontroller for processing. The versatility of male-to-male jumper wires extended beyond simple connections, allowing for easy modifications and debugging by quickly moving or replacing wires without significant disruption. Additionally, their color-coded insulation helped organize and identify different connections, reducing assembly errors and making the development process more efficient.

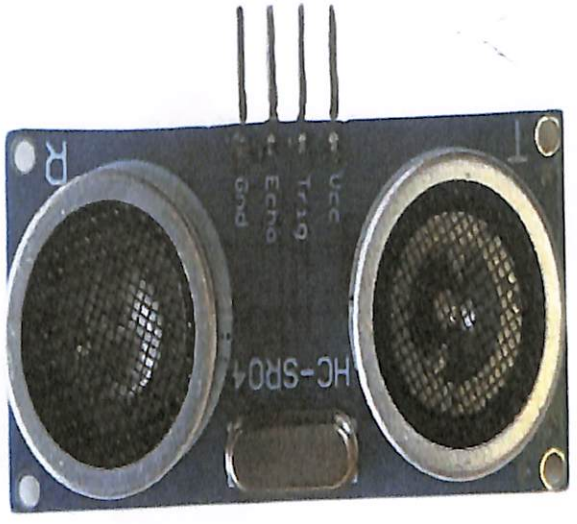


Plate 5. Ultrasonic Ranging Sensor

Plate 5 displays the ultrasonic ranging sensor, commonly known as an ultrasonic distance sensor, which functioned by emitting ultrasonic waves and calculating the time taken for the waves to return after bouncing off an object. In a flood alert system, this sensor was instrumental in monitoring water levels, accurately determining water depth or the proximity of water to critical levels. As the water rose, the sensor detected changes in distance and triggered alerts once predefined thresholds were crossed. Integrated with a central processing unit like an Arduino, the sensor sent real-time data to activate alert mechanisms such as SMS notifications. These alerts were crucial for informing both the head of the Municipal Disaster Risk Reduction and Management Office (MDRRMO) in the Municipality of Santo Tomas and residents of Barangay Narvacan, Sto. Tomas, La Union, enabling timely evacuation and preparation.

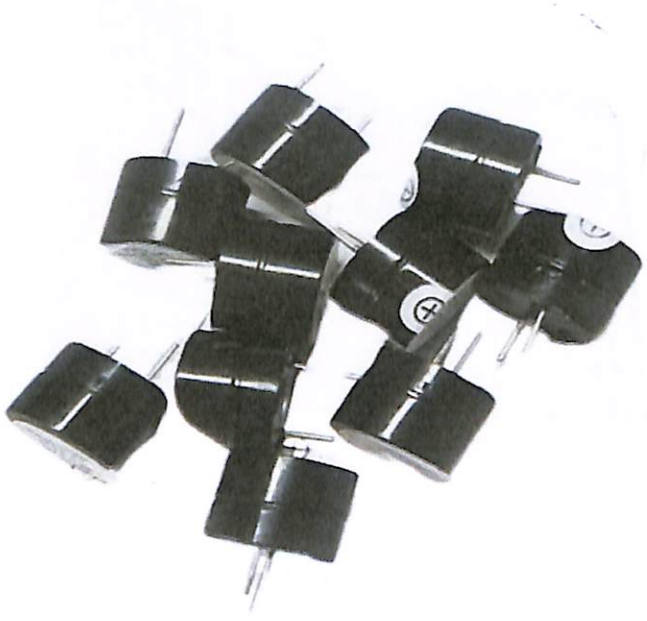


Plate 6. 12V Active Buzzer

Plate 6 presents the active buzzer, which served as an integral component for producing audible alerts in a flood alert system. Upon detection of a potential flood, the system could activate the buzzer to promptly warn nearby residents or authorities. The buzzer's 12V rating ensured that it emitted loud enough alerts to be heard effectively even in noisy environments, enhancing its effectiveness as an alarm device. This audible alert capability provided a crucial layer of notification in flood-prone areas, supplementing other communication methods and ensuring that individuals without access to digital alerts could still receive timely warnings. Additionally, the use of an active buzzer allowed for quick and direct activation by the system's central processing unit, enabling rapid response to changing flood conditions and potentially mitigating the impact of disasters.

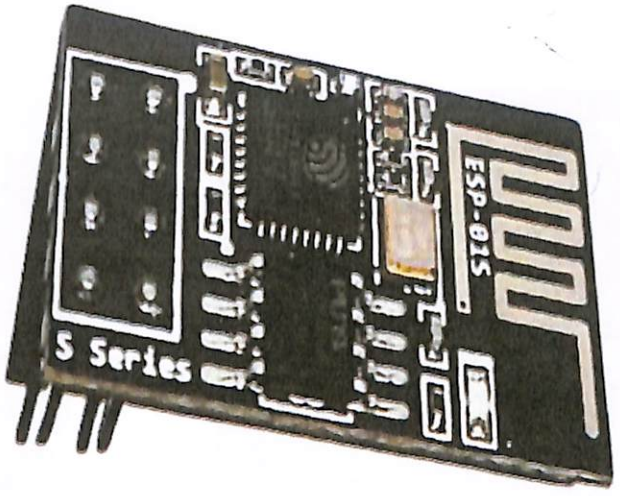


Plate 7. ESP-01S WiFi Module

Plate 7 illustrates the ESP-01S WiFi module, which played a crucial role as a component in an Arduino-based IoT Flood Alert System. By connecting the Arduino to the internet via WiFi, it facilitated the transmission of water level data from sensors in real-time. This capability enabled the Arduino to send flood alerts containing accurate water level information directly to smartphones or computers, offering timely warnings to residents and authorities. Such alerts were invaluable in providing early detection and response to potential flood threats, potentially saving lives and minimizing property damage. The integration of the ESP-01S WiFi module exemplified the power of IoT technology in enhancing disaster preparedness and response efforts, demonstrating how connectivity and data transmission could be leveraged to mitigate the impact of natural disasters.

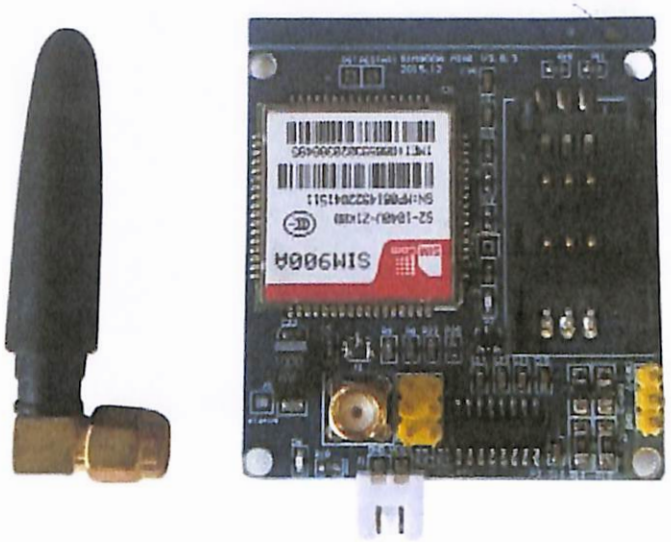


Plate 8. GSM Module SIM900A

Plate 8 displays the SIM 900A GSM module, facilitating communication over cellular networks within a flood alert system. By leveraging this module, the system could send SMS messages or make calls to notify relevant authorities or residents about potential flooding, ensuring timely alerts. Additionally, the GSM module could send SMS alerts upon detecting an increase in water level, providing critical real-time information. However, for the module to function correctly, it was essential to have a valid SIM card and an active mobile network subscription. This requirement ensured seamless connectivity and reliable communication, enabling the flood alert system to effectively fulfill its purpose of providing early warnings and enhancing disaster preparedness and response efforts.

Plate 9. 18650 Battery 3.7v Li-Ion

Plate 9 presents the 18650 3.7V Li-Ion battery, which was essential in an IoT-based flood alert system, providing reliable power to various components including the GSM Module SIM900A, ESP-01S WiFi Module, Ultrasonic Ranging Sensor, Circuit Rocks Breadboard 400, and Arduino Uno R3. This battery's high capacity and stable voltage ensured continuous operation of the GSM module for sending alert notifications, the WiFi module for network connectivity, and the ultrasonic sensor for accurate water level measurements. The Arduino Uno R3 orchestrated these components, processing data and triggering alerts, while the breadboard allowed for efficient and flexible circuit assembly. The 18650 battery's longevity and rechargeability made it ideal for sustaining the system in remote or power outage-prone areas, ensuring uninterrupted flood monitoring and timely alerts.

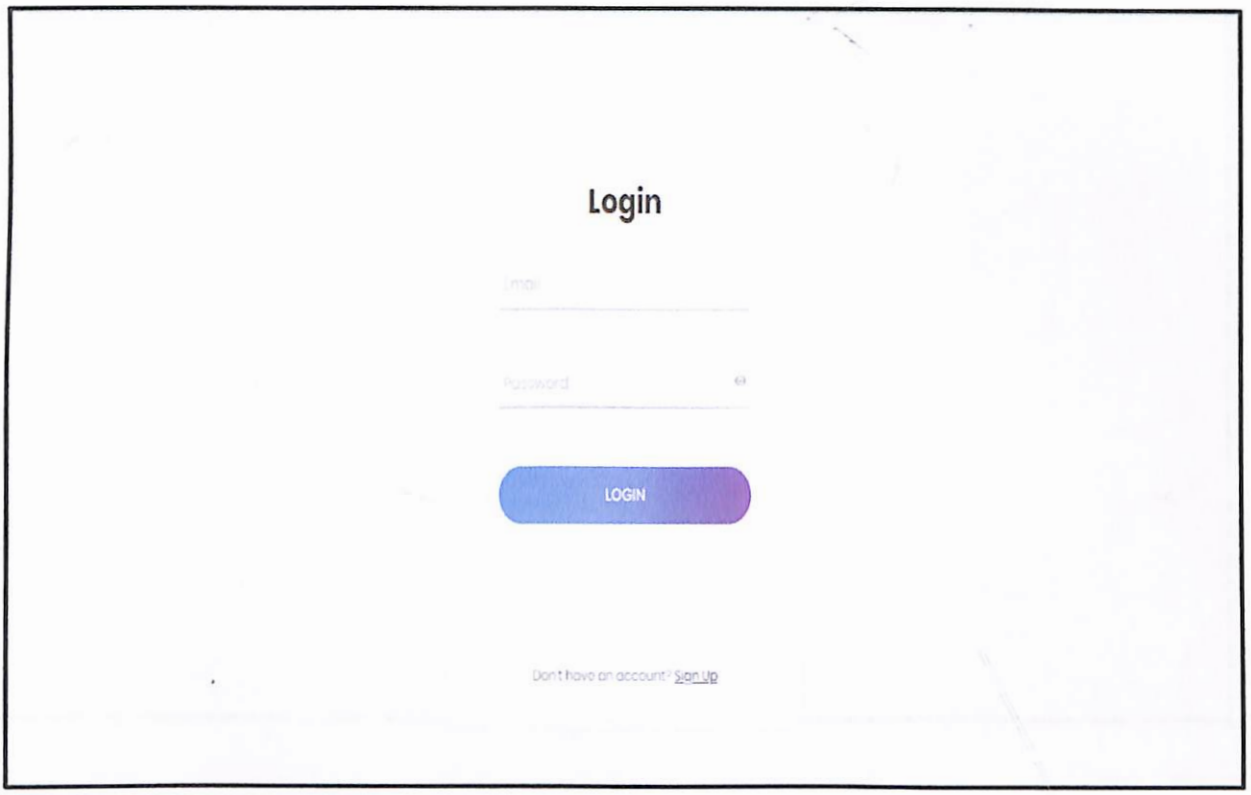


Plate 10. Login Page

Plate 10 presents the login screen, serving as the secure entry point for users to access the system's features. This critical screen features email and password fields, ensuring user identity verification through safe and encrypted credentials. The inclusion of a "Forgot Password" feature further enhanced security, providing users with an easily accessible means to retrieve access in case of forgotten passwords. Beyond its basic function, the login page serves as the system's initial line of defense, implementing strict security protocols to safeguard against unauthorized access. By enforcing robust authentication measures, the system-maintained integrity and protected sensitive data from potential threats. Additionally, user-friendly features such as the "Forgot Password" option contributed to a seamless user experience, reinforcing trust and usability within the system.

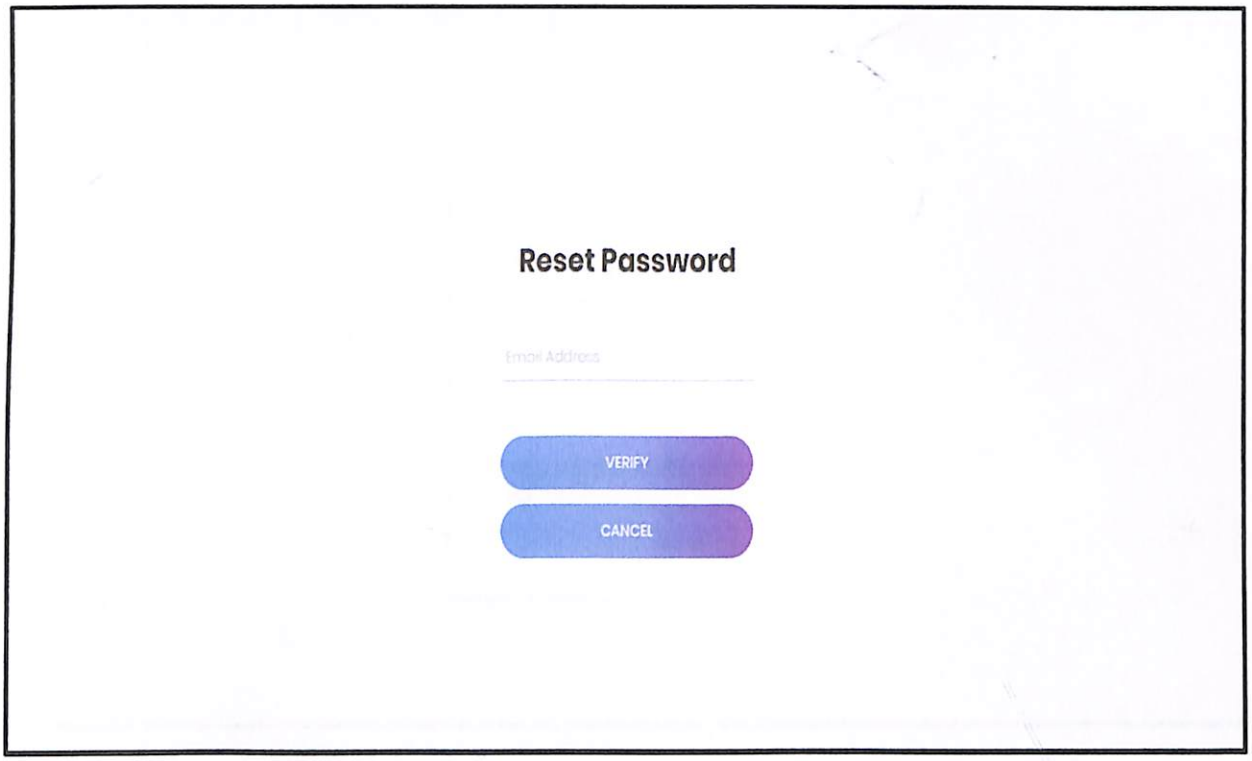


Plate 11. Reset Password Page

Plate 11 displays the reset password interface, designed to aid users who had forgotten their passwords and needed to regain access to their accounts. This feature comprises two primary sections. The first section features an email address field, prompting users to input the email address associated with their account. The second section includes a "Verify" button, which, when clicked, initiated the password reset process. Additionally, a "Cancel" button may have been available, providing users with the option to abort the password reset attempt if necessary. This user-friendly interface not only facilitates the password recovery process but also ensures a seamless experience for users encountering login difficulties. By offering clear instructions and intuitive navigation, the reset password features enhanced usability and user satisfaction within the system.

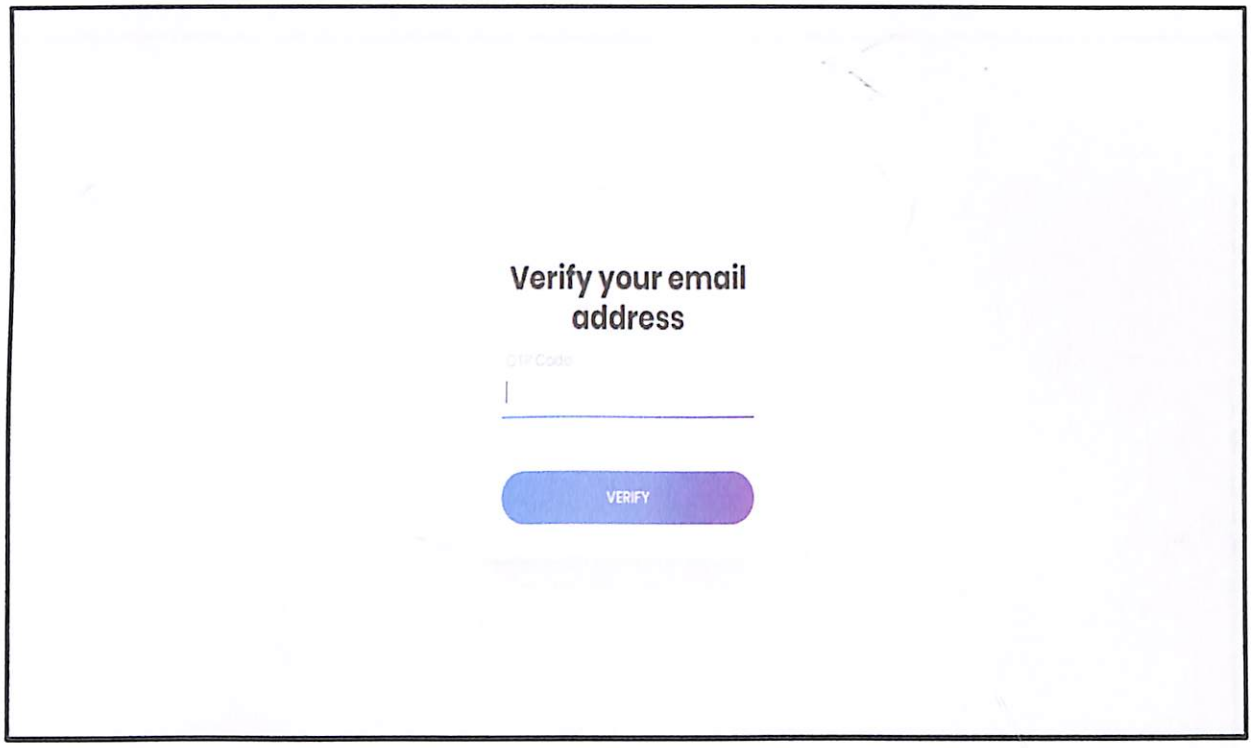


Plate 12. Verify Email Address

Plate 12 displays a website's past interaction with a user, guiding them through email verification as a security measure to ensure only authorized users accessed their accounts. The process was straightforward: the user entered their email address, triggering the generation and dispatch of a unique, temporary code, likely a One-Time Password (OTP), to that email. To complete verification, the user retrieved the code and entered it in the designated "OTP Code" field. Clicking "Verify" then confirmed their identity and granted access to their account. This two-step verification mechanism served as a robust safeguard against unauthorized access, adding an additional layer of security by requiring users to prove ownership of the email associated with their account.

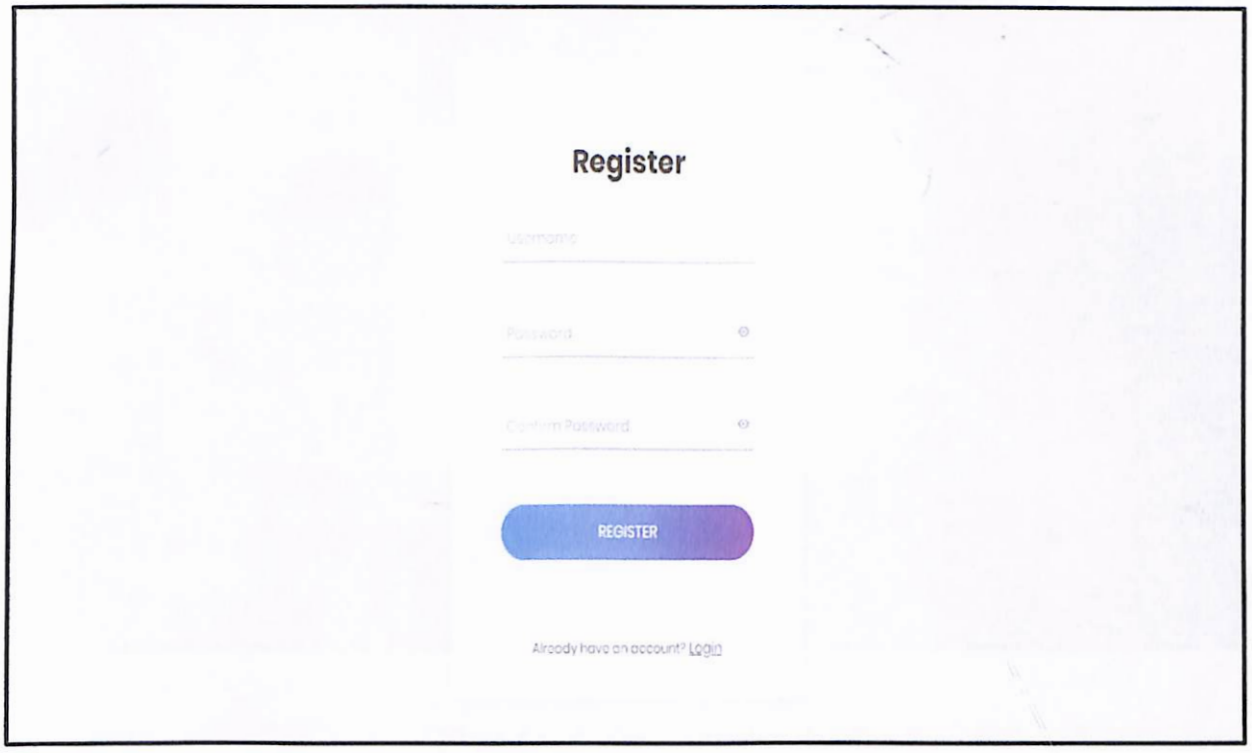


Plate 13. Account Registration Page

Plate 13 illustrates the account registration page, enabling users to create an account and access the system's features. The email address and password fields were pivotal for providing accurate information during the registration process. Once registered, users could quickly log in by clicking the "Sign In" button to access all the system's functionalities. Upon entering their information, the login button triggered a pop-up message confirming "Login Successful!". This streamlined login process enhanced user interaction with the system, offering immediate reassurance of access and reinforcing user confidence in the system's functionality.

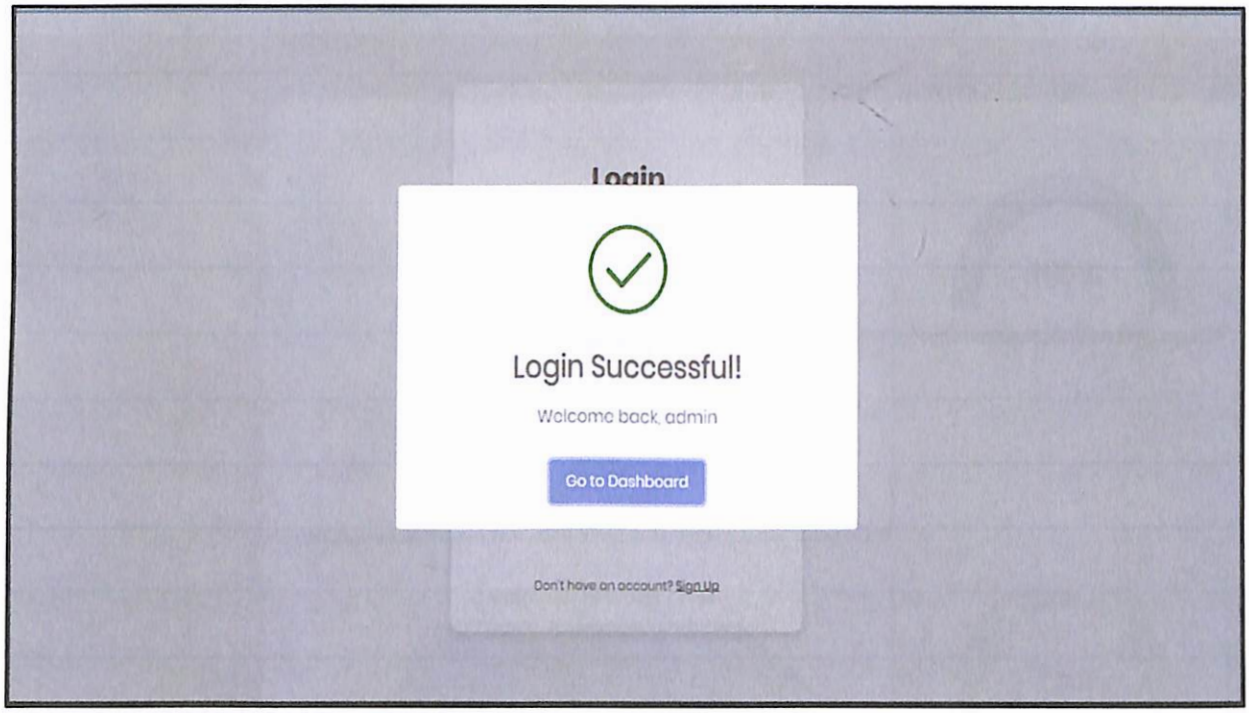


Plate 14. Successfully Login

Plate 14 indicates a button labeled "Go to Dashboard" appeared, allowing the admin to seamlessly navigate. This streamlined process facilitated user engagement and encouraged administrators to explore the system's dashboard promptly. By incorporating intuitive design elements and clear feedback mechanisms, the account registration page enhanced user experience and fostered a sense of efficiency and satisfaction within the system. The seamless transition to the dashboard emphasized user-centric design principles, ensuring that administrators could quickly access and utilize the system's core functionalities without encountering unnecessary obstacles or delays. However, the user-friendly interface and intuitive navigation contributed to a positive user experience, promoting user adoption and satisfaction with the system.

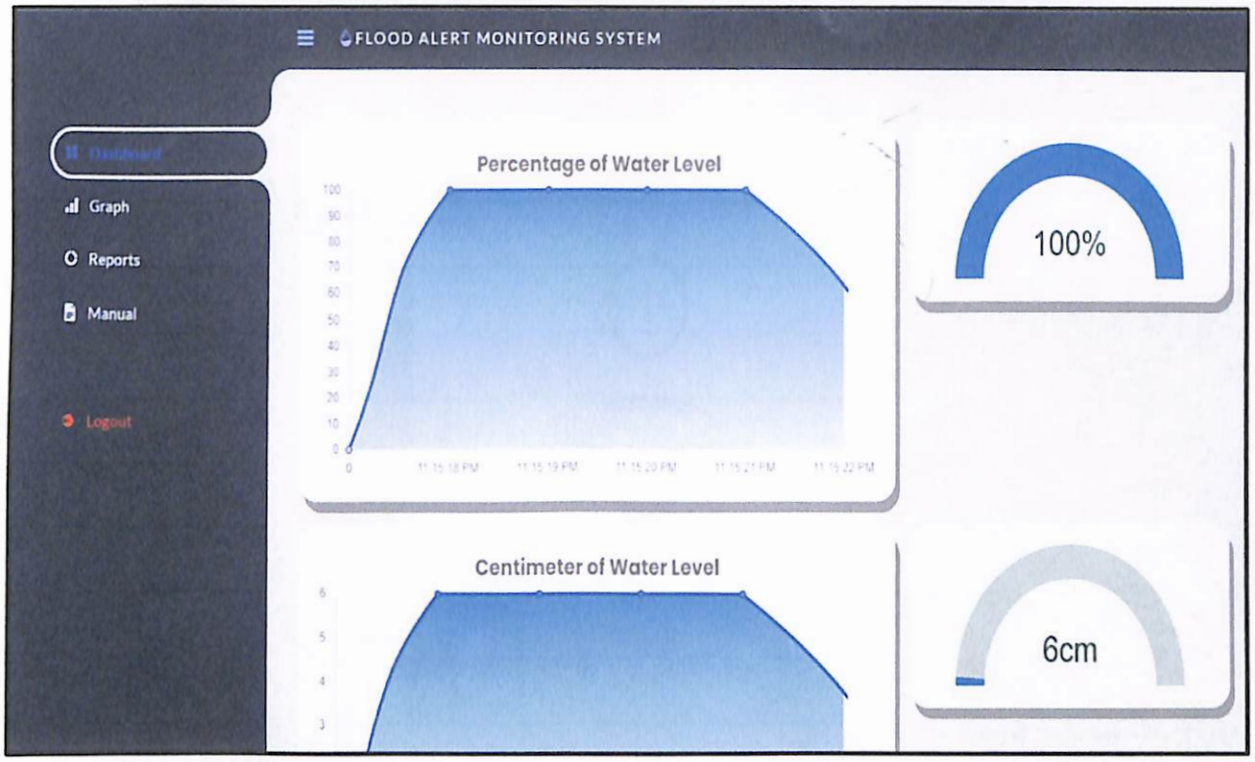


Plate 15. Dashboard Page

Plate 15 displays the user's dashboard which typically displays comprehensive flood alert monitoring system. It showcases real-time water level data through a digital display, indicating the levels in both centimeters and percentages. The interface is visually appealing and features a designated button that provides access to a graph that displays historical water level trends. This graph visually represents the fluctuating water levels, allowing users to monitor the situation effectively. The dashboard includes additional functionalities, including detailed reports and a manual input option, which can be accessed through buttons labeled "Reports" and "Manual," respectively. Finally, a log-out button is included so users can sign out of the system. Overall, this dashboard provides a clear snapshot of current water levels and offers options to delve deeper into historical data and even input manual readings.

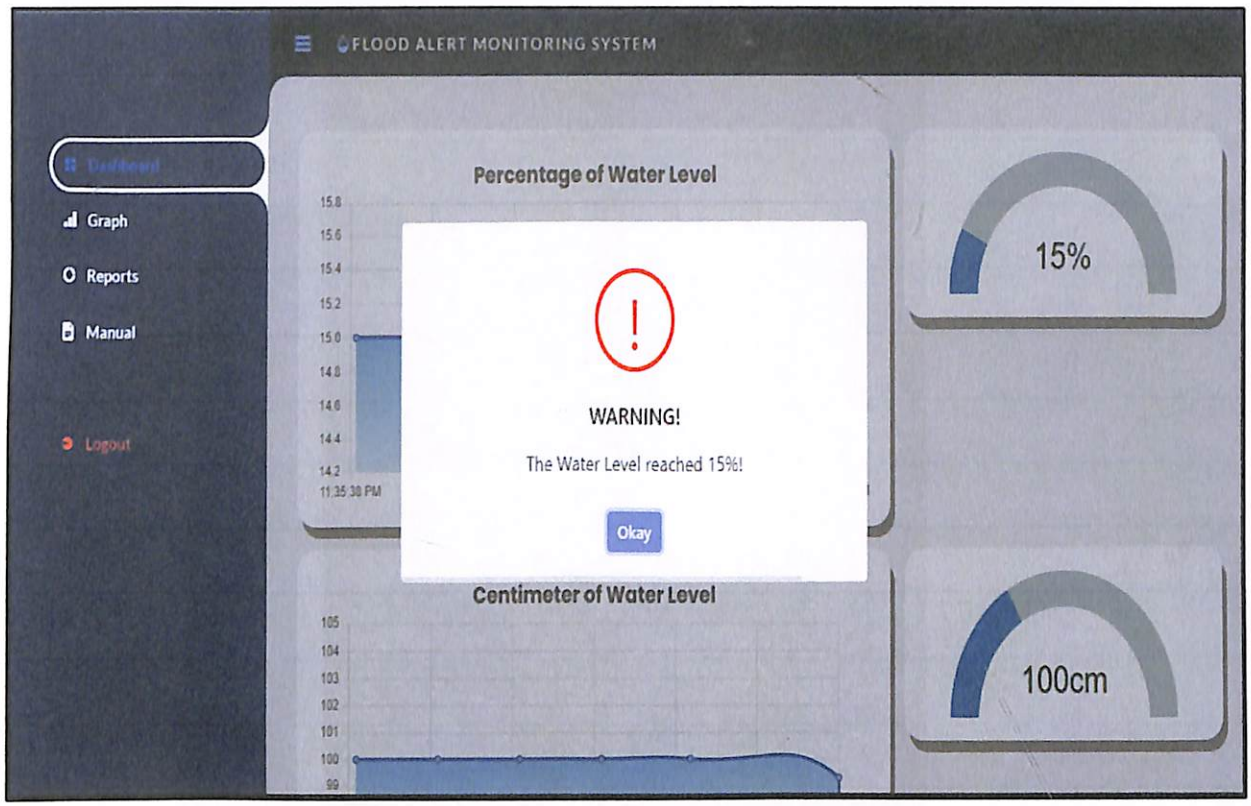


Plate 16. Notification Alert

Plate 16 presents the flood alert system, equipped with a range of notification methods, including pop-up messages from Alert Level 1 to Alert Level 3, alert sounds, and attention-grabbing visual indicators. These features enabled the system to swiftly detect critical flood situations or significant changes in water levels. By providing timely and multi-faceted alerts, the system kept users informed, empowering them to take necessary actions and precautions in real-time and ensuring their safety and preparedness. This comprehensive approach to flood monitoring and notification underscored the system's effectiveness in enhancing community resilience and mitigating the impact of floods on vulnerable areas.

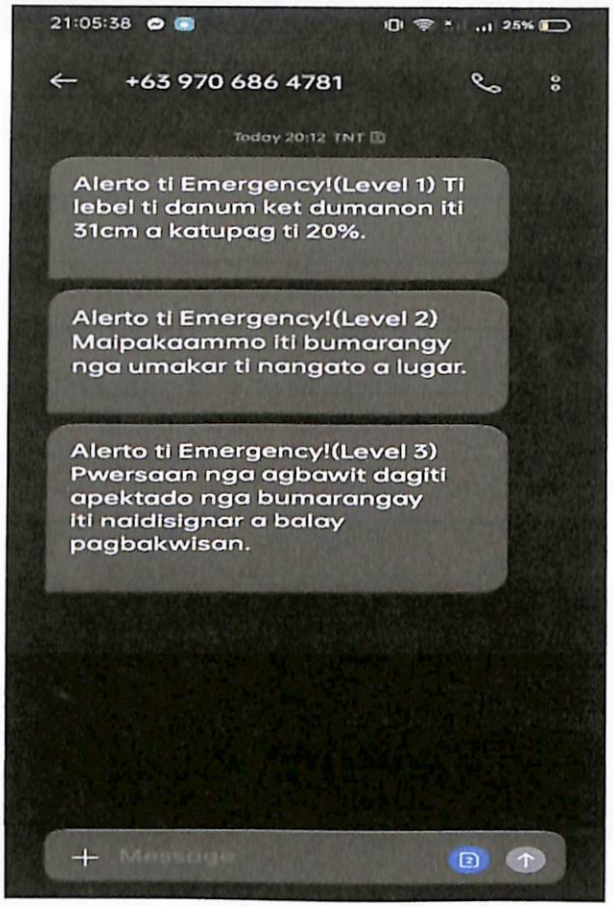


Plate 17. SMS Notifications

Plate 17 presents the SMS notifications, equipped with a range of notification methods, including pop-up messages from Alert Level 1 to Alert Level 3. The alert levels were clearly defined: Level 1 warned with the exact measurement of water level and its equivalent percentage, Level 2 instructed residents to move to higher ground, and Level 3 mandated evacuation, especially in affected areas. This comprehensive approach ensured timely communication of flood risks and appropriate response actions to users. Furthermore, the integration of these notification methods showcased the system's adaptability and responsiveness to evolving user needs, contributing to its overall functionality and usability in flood alert scenarios.

Plate 18. Data Analytic Graph Page

Plate 18 showcases the users that can obtain crucial insights about flood conditions through a data analytic graph. The graph page delivers a comprehensive visualization of water level fluctuations, featuring advanced analytical tools that pinpoint trends and patterns in the data. The graph updates in real time, reflecting any changes the flood alert monitoring system detects. With this integration, users gain a more comprehensive perspective of the flood situation, enabling them to make informed decisions based on data. Ultimately, this integration enhances their capacity to monitor, analyze, and respond to flood events more effectively.

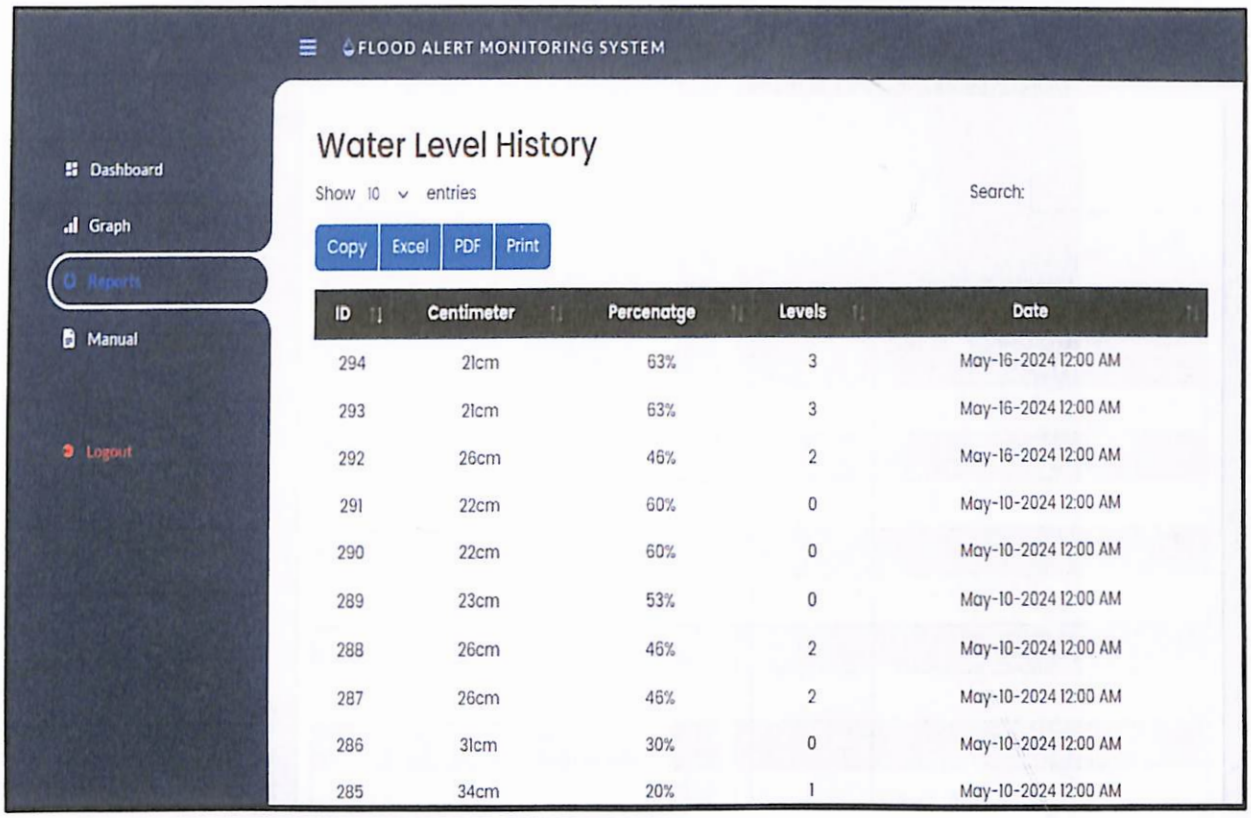


Plate 19. History Report Page

Plate 19 displays the monthly water history report which is invaluable for tracking and analyzing water data. Its well-crafted design offers a comprehensive overview of water levels over time, complete with various data points, charts, and graphs that enable users to visualize water-level trends and patterns. Presented historical water levels clearly and organized, the report allowed users to gain insights into changes, fluctuations, or any notable occurrences related to water levels. This feature ensured that users could easily access and share the report for analysis, record-keeping, or communication purposes. The water history report was an exceptional resource for understanding and monitoring water levels over time. This report allowed users to confidently track and analyze water data to make informed decisions and take necessary actions.

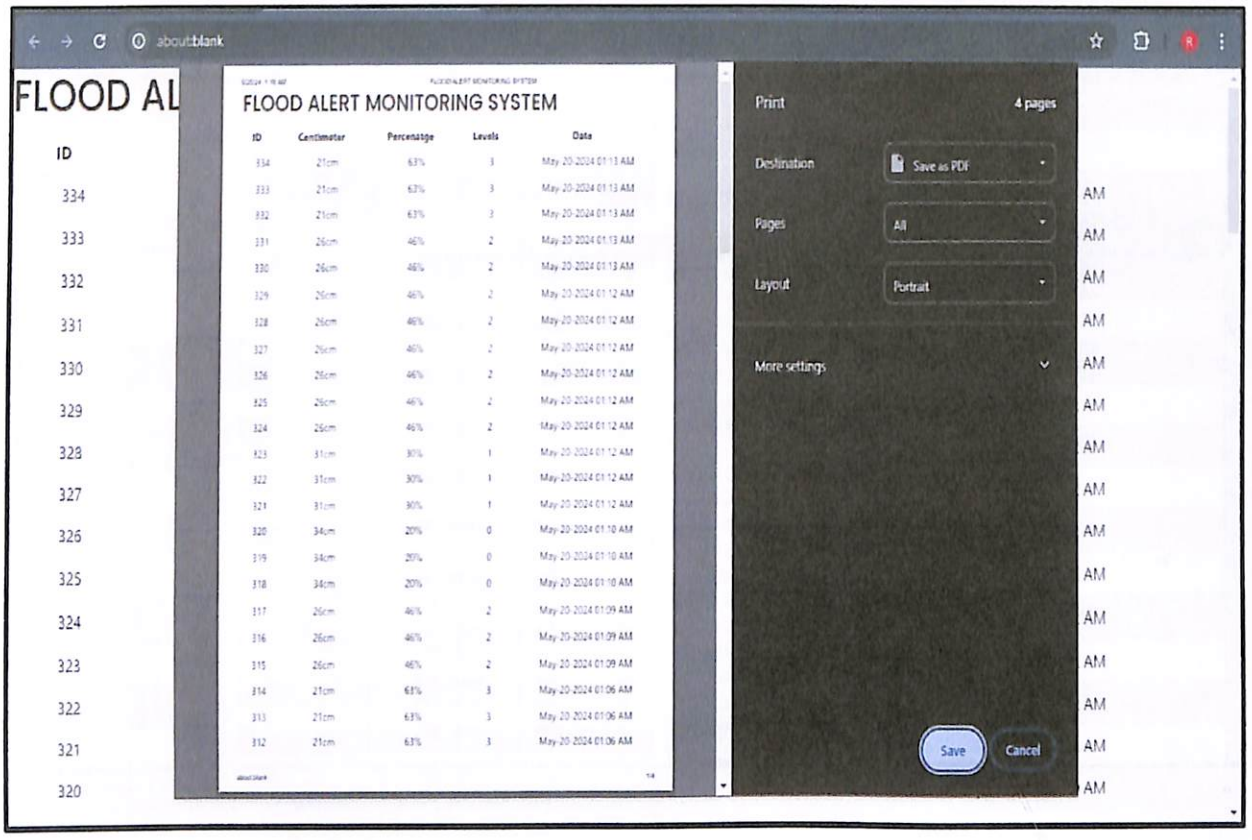


Plate 20. Sample Print Generated Report

Plate 20 illustrates that Reports often included a "Print" function, even if there was no physical button on the screen. Clicking this digital button triggered an automated process. The report was formatted for printing and sent directly to the default printer, saving time and ensuring an accurate transfer to paper. This feature streamlined the printing process, eliminating the need for manual formatting and reducing the risk of errors. Additionally, by directly sending the report to the default printer, users were spared the hassle of selecting a printer each time they wanted to print a report. Moreover, the inclusion of the "Print" function in digital form enhanced the efficiency and convenience of generating hard copies of reports within the system.

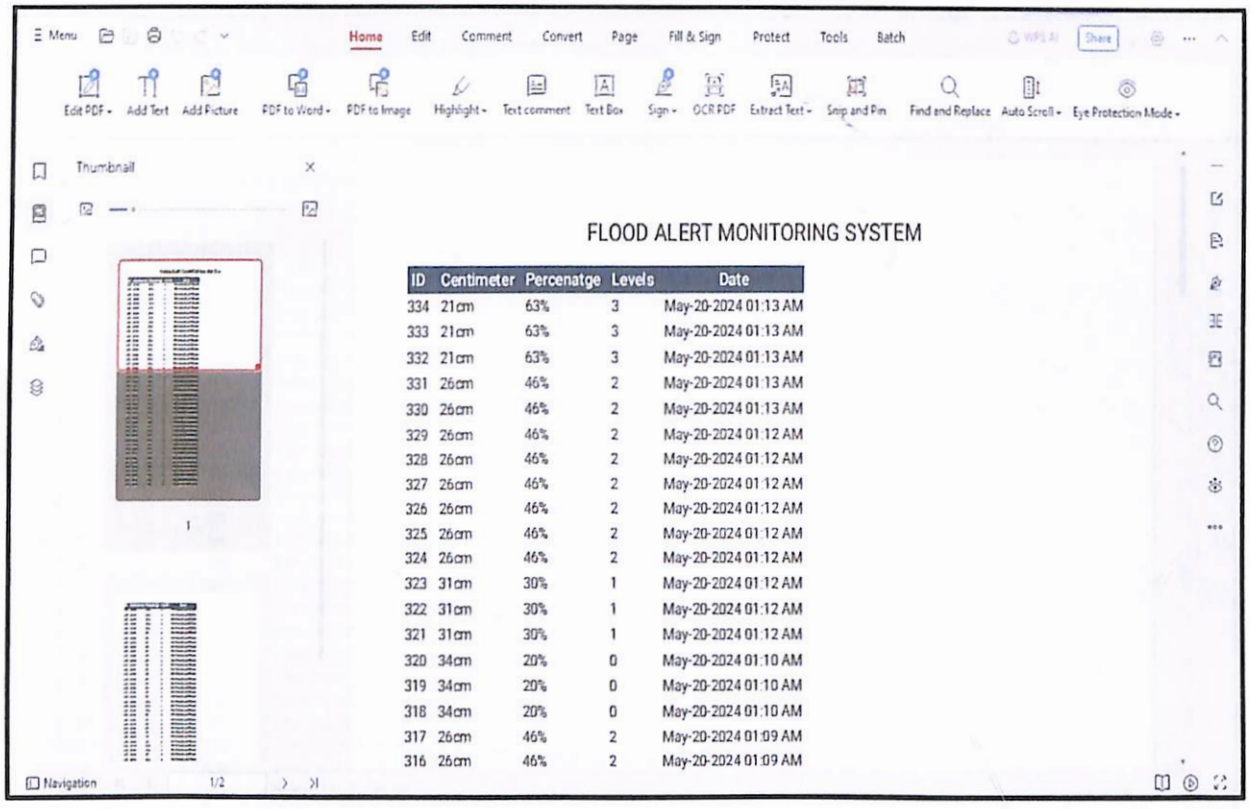
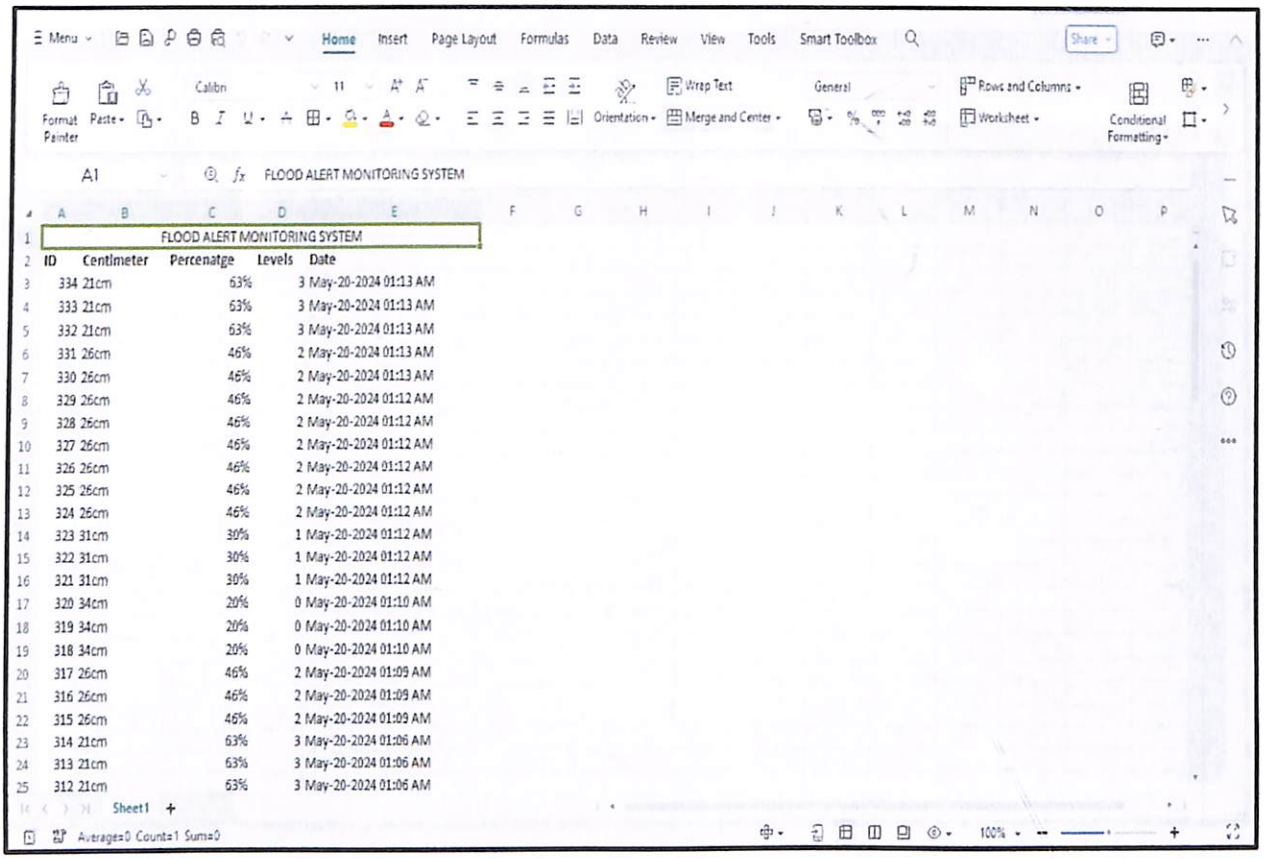


Plate 21. Sample PDF Generated Report

Plate 21 indicates that many system reports offered a "Download" option alongside the "Print" function. Clicking this digital button initiated a process that converted the report into a PDF file, ensuring the layout and formatting remained consistent across different devices. This feature provided users with flexibility in accessing and storing reports, as PDF files were widely compatible and could be easily shared or archived for future reference. Once the conversion was complete, users were prompted to save the PDF file to a chosen location on their device, enabling convenient access to the report at any time. This seamless integration of the "Download" option enhanced user experience by offering an efficient way to access and manage report data, promoting usability and accessibility within the system.



FLOOD ALERT MONITORING SYSTEM

ID	Centimeter	Percenatge	Levels	Date
334	21cm	63%	3	May-20-2024 01:13 AM
333	21cm	63%	3	May-20-2024 01:13 AM
332	21cm	63%	3	May-20-2024 01:13 AM
331	26cm	46%	2	May-20-2024 01:13 AM
330	26cm	46%	2	May-20-2024 01:13 AM
329	26cm	46%	2	May-20-2024 01:12 AM
328	26cm	46%	2	May-20-2024 01:12 AM
327	26cm	46%	2	May-20-2024 01:12 AM
326	26cm	46%	2	May-20-2024 01:12 AM
325	26cm	46%	2	May-20-2024 01:12 AM
324	26cm	46%	2	May-20-2024 01:12 AM
323	31cm	30%	1	May-20-2024 01:12 AM
322	31cm	30%	1	May-20-2024 01:12 AM
321	31cm	30%	1	May-20-2024 01:12 AM
320	34cm	20%	0	May-20-2024 01:10 AM
319	34cm	20%	0	May-20-2024 01:10 AM
318	34cm	20%	0	May-20-2024 01:10 AM
317	26cm	46%	2	May-20-2024 01:09 AM
316	26cm	46%	2	May-20-2024 01:09 AM
315	26cm	46%	2	May-20-2024 01:09 AM
314	21cm	63%	3	May-20-2024 01:06 AM
313	21cm	63%	3	May-20-2024 01:06 AM
312	21cm	63%	3	May-20-2024 01:06 AM

Plate 22. Sample Excel Generated Report

Plate 22 shows that some reports might also have offered a "Download (Excel)" option. Selecting this digital button triggered a conversion process where the report data was transformed into a well-organized Excel spreadsheet. This format enabled easy manipulation, analysis, and integration with other spreadsheets, enhancing the usability and versatility of the report data. The inclusion of both PDF and Excel download options provided users with flexibility in accessing and utilizing report information according to their specific needs and preferences. Whether users preferred a static, universally accessible format like PDF or a dynamic, data-rich format like Excel, they could easily obtain the report in their preferred format, allowing for seamless data management and analysis within the system.

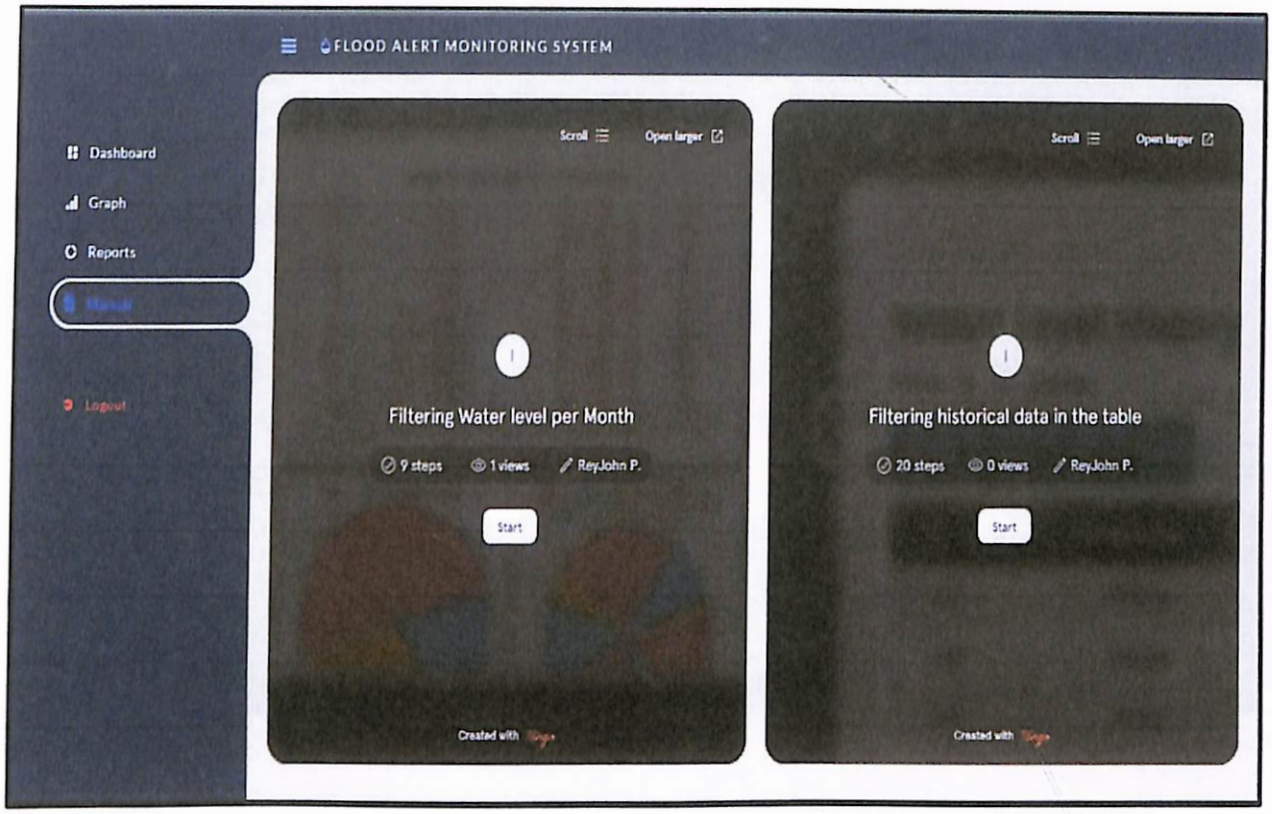


Plate 23. Manual Page

Plate 23 illustrates that the system remained incomplete in the absence of manuals. It featured two manuals alongside a "Start" button, enabling users to access or view the system's manual upon clicking. These manuals served as essential resources for users, providing comprehensive guidance on system functionalities, usage instructions, troubleshooting tips, and other relevant information. By offering easy access to manuals directly within the system interface, users could quickly reference and navigate through the documentation as needed, enhancing their understanding and proficiency in utilizing the system effectively. The inclusion of manuals not only facilitated user training and onboarding but also promoted self-service support, reducing the need for external assistance and empowering users to resolve issues independently.

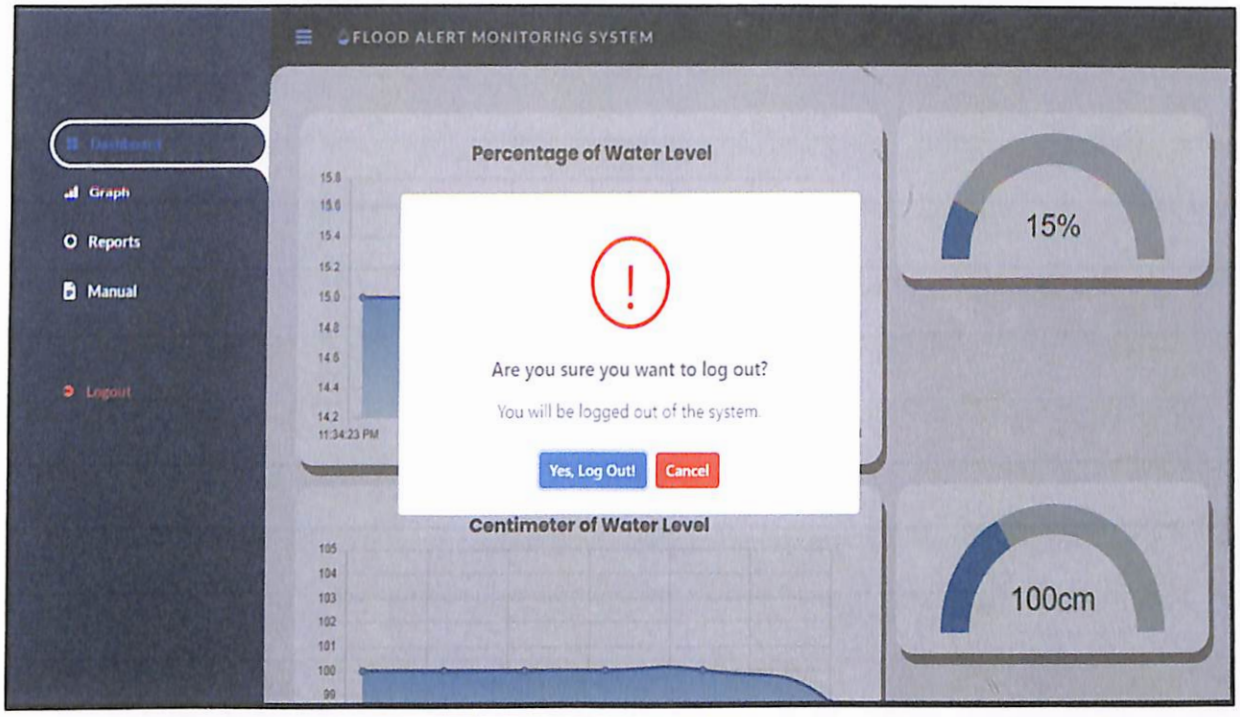


Plate 24. Logout Page

Plate 24 showcases that the alert box in the image was a confirmation prompt asking the user if they were sure they wanted to log out. This type of confirmation dialog was a common security measure to prevent accidental logouts. The box contained two buttons: "Yes, Log Out!" and "Cancel". The text in the alert box read: "Are you sure you want to log out? You were logged out of the system." This message clearly informed the user of the action they were about to take and its consequences (see Plate 25 on page 74). If users were unsure about the consequences of logging out, it was best to err on the side of caution and click "Cancel". They could always log out later if they were sure, it was safe to do so. This simple yet effective confirmation dialog helped prevent unintended logouts and ensured that users could make informed decisions about their actions within the system, enhancing overall usability and security.

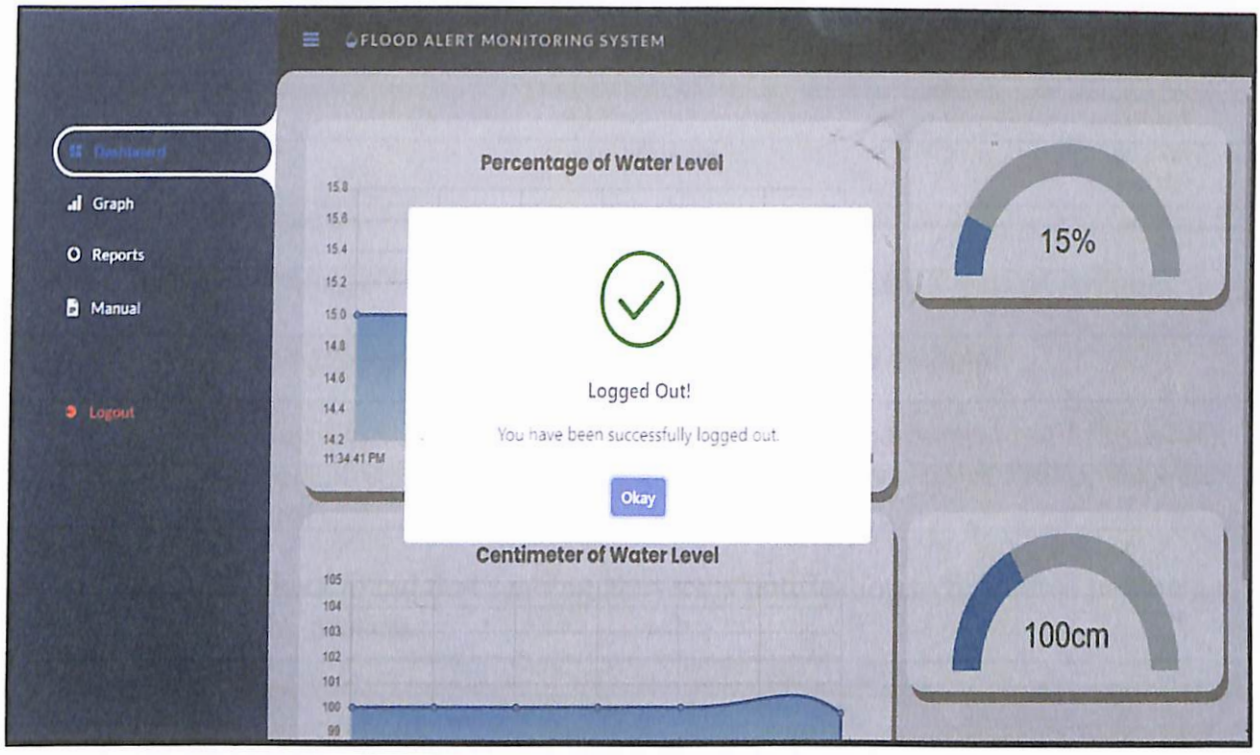


Plate 25. Successfully Logged Out

Test. After developing the prototype, the researchers undertook a comprehensive testing phase consisting of alpha and beta testing to ensure the system's quality and readiness for full-scale deployment. Alpha testing involved internal testing by the developers and IT experts within the development environment to identify and address major issues or bugs. This phase was crucial for uncovering and resolving critical flaws, performance bottlenecks, and usability concerns before exposing the system to external users. The meticulous documentation of errors encountered by researchers in Table 3 and Table 4 during alpha testing facilitated a systematic problem-solving approach, allowing for targeted improvements and bug fixes. Table 5 documented errors provided by external/end-users, respectively, during the beta testing phase.

Table 3. Errors Encountered by the Researchers in Alpha Testing

No.	Errors
01	Conflict between GSM module and ESP 01S for TX and RX pins of Arduino.
02	Unsuccessful attempts to allocate separate pins for each module.
03	When sending SMS alerts and notifications to locals and the head of MDRRMO, researchers adapted the local format rather than the conventional Philippine water level standard.
04	The researchers found that sending alert texts notification to registered numbers was a lengthy process.

Table 4. Errors Encountered by the IT Experts in Alpha Testing

No.	Errors
01	The microcontrollers and modules were not properly built in and were disorganized.
02	The report menu for the water level history showed multiple identical entries with no variations in the data.
03	In the dashboard, the use of colors to show level status (Warning/Danger).
04	The prototype had an adequate power supply from the battery.
05	Other components, such as transistors and resistors for circuit safety, were not included.

Errors identified during alpha testing highlighted areas for improvement. Resolving hardware issues like pin conflicts and disorganized components enabled proper functionality. Fixing reporting errors and color-coding enhanced accuracy and user

experience. Adapting to local standards and optimizing alerts ensured community alignment and timely notifications. By systematically fixing these errors, the team delivered a more robust and user-friendly flood alert system.

Table 5. Test Cases Performed by the Head of MDRRMO in Beta Testing

Test Case Type	Description	Test Step	Expected Result	Status
System Usefulness	Evaluate the ease of use and user-friendliness of the system's interfaces.	Provide end-users with tasks or scenarios involving monitoring sensor data and responding to alerts.	Interfaces should be intuitive and user-friendly, enabling easy navigation.	Passed
Information Quality	Evaluate system reliability under extreme conditions and failures.	Simulate different failure scenarios, such as sensor malfunctions, or system component failures.	The system should maintain accurate and reliable data, despite failures or interruptions.	Failed
	Verify flood detection and alerting based on sensor data.	Simulate a flood condition by increasing the water level.	When the water level exceeds the threshold, the system should trigger alerts like SMS notifications.	Failed
Interface Quality	Verify the system's security against unauthorized access.	Attempt to gain unauthorized access to the system's components.	The system should detect and mitigate security threats.	Failed

Continuation of Table 6

Test Case Type	Description	Test Step	Expected Result	Status
Interface Quality	The system's security against unauthorized access was verified.	The system's security against unauthorized access has been verified through OTP.	The system's security against unauthorized access was verified using OTP, ensuring secure access.	Failed
Overall System Testing	Verified ease of use, user-friendliness, informed decision safety for flood alerts, accuracy of flood detection, alerting, and system security.	End-users evaluated flood alert usability. Flood simulation conducted. System security verified with OTP.	User-friendly interfaces triggered alerts and notified residents and barangay officials of exceeded water levels, with security verified via OTP, passing the test.	Passed

In addition to the rigorous alpha and beta testing phases, the researchers presented the prototype to selected respondents within Barangay Narvacan, Sto. Tomas, La Union, to conduct system testing and gather feedback for further refinement. The primary objective was to identify any areas for improvement or potential issues that needed addressing in the real-world environment where the system would be deployed. Throughout this process, the researchers actively solicited comments and suggestions from the respondents, aiming to incorporate their perspectives into the system enhancement process.

After successful alpha testing, the software underwent beta testing, where it was released to a selected group of end-users (refer to Table 1 on page 25) to be used in their production environment. Beta testing aimed to thoroughly assess the software and hardware product's quality, functionality, and performance in a real-world setting. Utilizing the User Technology Acceptability Questionnaire, the IoT-Based Flood Alert System was rigorously examined, and any remaining defects were identified, ensuring that the product met the specified requirements and delivered value to the intended users before its final deployment. During the beta testing phase, the researchers closely monitored the system's performance and gathered valuable feedback from the end-users. This feedback played a crucial role in identifying potential issues, areas for improvement, and user experience challenges that may have been overlooked during the initial development stages. By involving the actual users in a real-world environment, the beta testing process provided insights into how the system would perform under various conditions and usage scenarios.

Deployment. After the testing of the system and satisfaction with the outcomes, the deployment phase of the agile methodology marked a critical step in implementing the developed system. Following thorough testing and confirmation that the system met the desired outcomes, it was ready for deployment to Barangay Narvacan, Sto. Tomas, La Union, and the Municipal Disaster Risk Reduction and Management Office (MDRRMO) of the Municipality of Santo Tomas. During this phase, it was crucial to facilitate a seamless transition from the testing environment to the live production environment.

This phase involved several key activities to ensure a smooth and successful implementation. The necessary infrastructure, such as setting up servers, configuring

networks, and ensuring reliable connectivity, was established to enable the system to function seamlessly and provide real-time flood alerts. The system was configured according to the specific requirements of the target areas, including customizing user interfaces, integrating with existing systems, and fine-tuning alert thresholds and notification mechanisms, which was crucial for alignment with local needs. Additionally, comprehensive training was provided to end-users, including local authorities, disaster management personnel, and community members, to ensure effective utilization of the system's capabilities. Engaging with stakeholders, including local authorities, community representatives, and disaster heads, was vital for gathering feedback and fine-tuning the system to align with their specific needs and ensuring their satisfaction with the final product.

The implementation team worked closely with local partners to understand the unique challenges and requirements of each target area. Rigorous testing and simulations were conducted to identify and address any potential issues or vulnerabilities before the system went live. Regular maintenance and update schedules were established to keep the system running optimally and incorporate any necessary improvements. A dedicated support team was set up to address any technical issues or user queries that might arise. Continuous monitoring and evaluation mechanisms were put in place to assess the system's performance and impact over time. Screenshots served as the representation of the IoT-Based flood alert system, as shown on the following pages. The researchers had recognized the importance of providing comprehensive support and maintaining open communication channels with end-users to ensure the successful adoption and sustained use of the system.

Additionally, the continuous monitoring and evaluation processes allowed for data-driven improvements and adaptations to be made, ensuring that the system remained effective and relevant in addressing the evolving needs of the community.

Review. After deployment and system implementation, it was not uncommon for errors or issues to have arisen during continued usage. While the Agile methodology emphasized iterative development and testing, the final phase did not always encompass ongoing maintenance and support.

Ongoing maintenance and support were essential aspects of any technology implementation, ensuring that the system continued to operate effectively and met the evolving needs of users. This included addressing software bugs, hardware failures, user support queries, and implementing updates or enhancements to improve system performance and functionality. However, due to the dynamic nature of technology and the unpredictability of user behaviors and external factors, ongoing maintenance and support could not always be fully covered within the timeframe of a research study.

Furthermore, the sustainability of the system relied heavily on the availability of resources, including funding, personnel, and infrastructure, beyond the duration of the research study. Establishing partnerships with local authorities, community organizations, or industry stakeholders could have helped ensure ongoing support and maintenance. Collaboration with these entities could also have facilitated the development of sustainable funding models and the establishment of mechanisms for user feedback and system improvement over time. Overall, while the Agile methodology provided a structured

approach to system development, the sustainability of technology implementations often required broader considerations and strategic planning beyond the initial project phases.

Acceptability Evaluation of the IoT-Based Flood Alert System

The utilization of the User Technology Acceptability Questionnaire (UTAQ) to measure the level of acceptability of the IoT-Based Flood Alert System for Narvacan, Sto. Tomas, La Union was a valuable approach. By administering the survey questionnaire to a significant number of respondents, which included 9 barangay council officials and 21 residents from various sectors, the researchers were able to gather thorough feedback on the proposed system. The researchers developed standards to improve system usefulness, information quality, and interface quality based on the results, thereby ensuring that the system effectively met the needs and expectations of its intended users. Through this iterative process of evaluation and refinement, the researchers addressed potential issues and optimized the system for maximum effectiveness in supporting flood preparedness and response efforts in the community.

Furthermore, the inclusion of diverse stakeholders, including barangay council officials and residents from various sectors, provided a comprehensive understanding of the system's acceptability from multiple perspectives. This inclusive approach helped to address potential concerns or barriers to adoption, ensuring that the system was accessible, user-friendly, and capable of effectively disseminating critical information during flood events. By iteratively refining the system based on user feedback, the researchers increased the likelihood of successful adoption and sustained use within the community, ultimately strengthening flood preparedness and response capabilities.

Based on the results shown in Table 7, it is evident that respondents assessed the developed system's usefulness with a high level of acceptability. Indicator 1 emerged as the most positively rated, with a mean score of 6.93, while the second indicator received a slightly lower mean score of 6.77. This indicates that users find the system quite intuitive and easy to navigate. The overall computed Average Weighted Mean of 6.87 further reinforced the system's remarkable acceptance among users. Such a high rating implied that the system was not only easy to use but also highly user-friendly, meeting the needs and expectations of its intended users effectively. This level of user satisfaction was a testament to the system's well-designed interface and intuitive functionality, which likely contributed to its successful adoption and widespread utilization at the time.

The system's exceptional performance was particularly noteworthy in the context of emergency response, where efficient and effective communication and monitoring could have life-saving implications. As highlighted by Gacu et al. (2022), such systems played a crucial role in improving emergency response times by enabling accurate and timely monitoring of individuals in need of assistance. By streamlining the process of identifying and locating those requiring immediate aid, the developed system contributed significantly to enhancing overall emergency management strategies. Consequently, the high level of user acceptance and satisfaction with the system's usefulness held profound implications for the field of emergency response, potentially translating into more lives saved and improved public safety outcomes.

Table 7. Level of Acceptability of the IoT-Based Flood Alert System in terms of System Usefulness

Indicators	Mean	Descriptive Equivalent Rating	Descriptive Interpretation
1. How often do you use the system to check for flood alerts?	6.93	Strongly Agree	Extremely High Acceptance
2. Do you find the system helpful in making informed decisions about your safety during flood events?	6.77	Strongly Agree	Extremely High Acceptance
3. Does the system provide enough information to prepare for floods effectively?	6.87	Strongly Agree	Extremely High Acceptance
4. Has the system helped you feel more secure and prepared for flood emergencies?	6.87	Strongly Agree	Extremely High Acceptance
5. Do you find the system a valuable tool for managing flood risks in your community?	6.90	Strongly Agree	Extremely High Acceptance
6. Overall, how satisfied are you with the usefulness of the system?	6.87	Strongly Agree	Extremely High Acceptance
Weighted Mean	6.87	Strongly Agree	Extremely High Acceptance

Legend:

Statistical Range	*Descriptive Equivalent Rating*	*Descriptive Interpretation*
6.16-7.00	*Strongly Agree*	*Extremely High Acceptance*
5.30-6.15	*Agree*	*Very High Acceptance*
4.44-5.29	*Somewhat Agree*	*Moderately High Acceptance*
3.58-4.43	*Neutral*	*Neutral Acceptance*
2.72-3.57	*Somewhat Disagree*	*Moderately Low Acceptance*
1.86-2.71	*Disagree*	*Low Acceptance*
1.00-1.85	*Strongly Disagree*	*Extremely Low Acceptance*

Table 8 presents an assessment of the level of acceptability regarding information quality within the developed system. This assessment provided valuable insights into the system's strengths and areas for potential improvement. Indicator number 9 emerged with the highest mean score of 6.90, indicating an extremely high acceptance level of performance. This suggested that the system excelled in presenting information in a clear, concise, and easily understandable manner, enabling users to comprehend and utilize the provided data effectively. On the other hand, indicator number 11 obtained the lowest mean score of 6.77, which, although still reflecting a high acceptance level, may have indicated opportunities for improvement in how information was structured and presented within the system.

The overall calculated Average Weighted Mean of 6.83 implied that the information provided by the system was well-structured and arranged at an extremely high acceptance level. This aligned with the findings of research conducted by Hu et al. (2022), which emphasized the importance of data preparation, assessment report generation, and efficient integrated visualization for decision-ready information. By addressing the potential areas for improvement identified in indicator number 11, the researchers could further enhance the system's ability to convey information in a structured and intuitive manner, ultimately increasing its effectiveness and acceptability among end-users.

Table 8. Level of Acceptability of the IoT-Based Flood Alert System in terms of Information Quality

Indicators	Mean	Descriptive Equivalent Rating	Descriptive Interpretation
7. Do you believe the information provided by the system is accurate and reliable?	6.83	Strongly Agree	Extremely High Acceptance
8. Is the information provided by the system timely and up-to-date?	6.80	Strongly Agree	Extremely High Acceptance
9. Do you find the information provided by the system to be clear and easy to understand?	6.90	Strongly Agree	Extremely High Acceptance
10. Does the information the system provides address your community's specific needs?	6.80	Strongly Agree	Extremely High Acceptance
11. How confident are you in the accuracy of the information provided by the system?	6.77	Strongly Agree	Extremely High Acceptance
12. Overall, how satisfied are you with the system's quality of information?	6.87	Strongly Agree	Extremely High Acceptance
Weighted Mean	6.83	Strongly Agree	Extremely High Acceptance

Legend:

Statistical Range	*Descriptive Equivalent Rating*	*Descriptive Interpretation*
6.16-7.00	*Strongly Agree*	*Extremely High Acceptance*
5.30-6.15	*Agree*	*Very High Acceptance*
4.44-5.29	*Somewhat Agree*	*Moderately High Acceptance*
3.58-4.43	*Neutral*	*Neutral Acceptance*
2.72-3.57	*Somewhat Disagree*	*Moderately Low Acceptance*
1.86-2.71	*Disagree*	*Low Acceptance*
1.00-1.85	*Strongly Disagree*	*Extremely Low Acceptance*

Table 9 shows the user's perception and acceptance of the developed system's interface, functions, and capabilities. The findings revealed both strengths and potential areas for further improvement. Indicator number 13 achieved the highest mean score of 6.93, indicating that the functions and capabilities of the developed system were extremely well-received and highly accepted by the users. This strong positive response suggested that the system successfully delivered the intended functionalities and met the users' expectations, enabling them to effectively utilize the system for its intended purpose. On the other hand, indicator number 14 obtained the lowest mean score of 6.77, although still suggesting an extremely high acceptance level regarding the likelihood and satisfaction of using the system's interface. This slight deviation from the highest score may have indicated potential opportunities for enhancing the user interface's intuitiveness or ease of use.

The calculated Average Weighted Mean of 6.87 implied that the overall interface quality, functions, and capabilities of the developed system were perceived as extremely high by the users. This high level of acceptance supported the assertion made by the Philippines (2018) that the system significantly enhanced the accessibility of hazard information, presenting it in a user-friendly format that fostered a more informed and resilient community response. The high mean scores across all indicators highlighted the system's effectiveness in meeting user needs and expectations, suggesting that such a system could serve as a model for future developments in hazard information dissemination.

Table 9. Level of Acceptability of the IoT-Based Flood Alert System in terms of Interface Quality

Indicators	Mean	Descriptive Equivalent Rating	Descriptive Interpretation
13. Do you find the system easy to navigate and use?	6.93	Strongly Agree	Extremely High Acceptance
14. Do you find the system interface visually appealing and user-friendly?	6.77	Strongly Agree	Extremely High Acceptance
15. Is the system accessible to users with different levels of technical expertise?	6.87	Strongly Agree	Extremely High Acceptance
16. Does the system provide users with clear instructions and helpful feedback?	6.87	Strongly Agree	Extremely High Acceptance
17. How intuitive do you find the system's design and layout?	6.90	Strongly Agree	Extremely High Acceptance
18. Overall, how satisfied are you with the system interface?	6.87	Strongly Agree	Extremely High Acceptance
Weighted Mean	6.87	Strongly Agree	Extremely High Acceptance

Legend:

Statistical Range	*Descriptive Equivalent Rating*	*Descriptive Interpretation*
6.16-7.00	*Strongly Agree*	*Extremely High Acceptance*
5.30-6.15	*Agree*	*Very High Acceptance*
4.44-5.29	*Somewhat Agree*	*Moderately High Acceptance*
3.58-4.43	*Neutral*	*Neutral Acceptance*
2.72-3.57	*Somewhat Disagree*	*Moderately Low Acceptance*
1.86-2.71	*Disagree*	*Low Acceptance*
1.00-1.85	*Strongly Disagree*	*Extremely Low Acceptance*

Table 10 presents a summary of the developed system's acceptability level based on evaluations from barangay official councils and residents per sectors of Barangay Narvacan, Sto. Tomas, La Union. The table consisted of three indicators: system usefulness, information quality, and interface quality. The system usefulness indicator received the highest mean score of 6.87, indicating that the system provided clear and easy-to-use functionalities. The interface quality indicator also garnered a high mean score of 6.82, suggesting that the interface was pleasant and likable. The overall calculated Grand Mean of 6.87 implied that the developed system provided clear, effective, and easily accessible information with a likable and pleasant interface. This suggests that, despite minor areas for improvement, the developed system excelled in meeting user expectations and requirements, demonstrating its effectiveness in addressing the needs of the stakeholders.

With an overall grand mean of 6.87, the developed system achieved an exceptional level of acceptability among its target users. This score implied that the system not only provided clear, effective, and easily accessible information but also offered a likable and pleasant interface. While there may have been minor areas for improvement, the high grand mean score suggested that the system excelled in meeting user expectations and addressing the needs of stakeholders. The positive feedback from both barangay officials and residents across different sectors further reinforced the system's effectiveness and its potential to significantly contribute to the community's development and progress.

Table 10. Summary of Acceptability

Indicators	Mean	Descriptive Equivalent Rating	Descriptive Interpretation
System Usefulness	6.87	Strongly Agree	Extremely High Acceptance
Information Quality	6.83	Strongly Agree	Extremely High Acceptance
Interface Quality	6.82	Strongly Agree	Extremely High Acceptance
Grand Mean	6.87	Strongly Agree	Extremely High Acceptance

Legend:

Statistical Range	*Descriptive Rating*	*Descriptive Equivalent*
6.16-7.00	*Strongly Agree*	*Extremely High Acceptance*
5.30-6.15	*Agree*	*Very High Acceptance*
4.44-5.29	*Somewhat Agree*	*Moderately High Acceptance*
3.58-4.43	*Neutral*	*Neutral Acceptance*
2.72-3.57	*Somewhat Disagree*	*Moderately Low Acceptance*
1.86-2.71	*Disagree*	*Low Acceptance*
1.00-1.85	*Strongly Disagree*	*Extremely Low Acceptance*

Chapter 4

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

Summary

This research aimed to develop an IoT-Based Flood Alert System for Narvacan, Sto. Tomas, La Union. The motivation behind the study was the vulnerability of the area to flooding due to its low-lying topography and proximity to rivers and the sea. The system was designed to monitor water levels in real-time and provide timely alerts to residents through SMS, enabling them to evacuate or take necessary precautions in case of potential flood events.

The researchers employed descriptive and developmental research methods, utilizing Agile methodology to guide the system's development. The system components included Arduino Uno, ultrasonic sensors, ESP-01s WiFi Module, Active Buzzer, and a web-based interface including C/C++, HTML, CSS, JavaScript, PHP, and MySQL, for an intuitive user interface and efficient data management. The system was designed to analyze water levels, predict potential floods, and send SMS alerts to residents.

To assess the system's acceptability, the researchers administered the User Technology Acceptability Questionnaire (UTAQ) to a sample of 30 respondents, including 9 barangay council officials and 21 residents from each of the seven sectors of Barangay Narvacan, Sto. Tomas, La Union. The user survey for the IoT-Based Flood Alert System painted a very positive picture.

The following are the salient findings of the study:

1. Through an interview questionnaire, the researchers were able to ascertain the prototype techniques used in the development of the system, which enabled them to successfully develop an IoT-based flood alert system using the Agile methodology model.
2. The IoT-Based Flood Alert System received overwhelming endorsement from users, with a grand mean score of 6.87 (Strongly Agreed). This reflected extremely high acceptance across all categories: usefulness (6.87), information quality (6.83), and interface quality (6.82). These scores indicated users found the system both valuable and informative, with nearly 98% rating it as such, and user-friendly and easy to interact with, with nearly 97% agreement. This strong user perception suggested the system had high potential for successful community adoption.

Conclusions

Based on the findings of the study, the following conclusions were drawn:

1. The Agile Methodology proved suitable for the development of the IoT-Based Flood Alert System. This software development utilized components that were highly helpful and applicable in developing the prototype. A prototype IoT-Based Flood Alert System was deployed in Narvacan, Sto. Tomas, La Union, using IoT to gather real-time data on water levels and environmental factors. It aimed to offer timely flood warnings, empowering residents to safeguard lives

and property.

2. To assess community adoption potential, a User Technology Acceptability Questionnaire evaluated the system's usefulness, information quality, interface, and overall acceptability. Based on the gathered data, IoT-Based Flood Alert System for Narvacan, Sto. Tomas, La Union was extremely high acceptance.

Recommendations

Based on the findings and conclusions, the researchers recommended the following:

1. To implement the developed prototype, which was intended to be adapted easily in Narvacan, Sto. Tomas, La Union, and other places within the Municipality of Santo Tomas, was recommended.
2. The researcher suggests that the Agile Methodology could be a valuable tool for future researchers developing IoT-based flood alert systems.
3. The implementation of IoT-Based Flood Alert System may be extended to the flood prone areas in the Philippines such as targeting coastal and low-lying regions.
4. The User Technology Acceptability Questionnaire (UTAQ) is highly recommended to assess system's acceptability in future researchers.
5. To aid future researchers, it is recommended to implement hardware monitoring tools in prototype development.

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APPENDICES

APPENDIX A

Interview Guide Questionnaire

Don Mariano Marcos Memorial State University
South La Union Campus
COLLEGE OF COMPUTER SCIENCE
Agoo La Union

Name: _____

1. What methods, both manual and automated, are used to predict flooding in different areas?

_____

_____

2. What solutions and preparations has your organization put in place as part of its flood preparedness plan?

_____

_____

3. What strategies can affect areas and organizations employ to overcome this situation?

_____

_____

APPENDIX B

Sample Response Interview Guide Questionnaire

Don Mariano Marcos Memorial State University
South La Union Campus
COLLEGE OF COMPUTER SCIENCE
Agoo La Union

Name: RICARDO C. COLLADO

1. What methods, both manual and automated, are used to predict flooding in different areas?

WE DONT HAVE ANY AUTOMATED METHODS BUT WE USED THE CALENDAR - BASED METHOD TO TRACK TIDAL CYCLES.

2. What solutions and preparations has your organization put in place as part of its flood preparedness plan?

PART OF OUR PREPAREDNESS PLAN IS TO INFORM THEM WHEN TO EVACUATE AND CONDUCT SEMINAR TOGETHER WITH MDRRMO.

3. What strategies can affect areas and organizations employ to overcome this situation?

OUR STRATEGIES IS TO EMPLOY EARLY WARNINGS, IMPROVE LAND AND INFRASTRUCTURE, AND CONTINOUSLY EDUCATING COMMUNITY.

APPENDIX C

User Technology Acceptability Questionnaire

(Justinlawrence & Hudson, 2022)

Republic of the Philippines
DON MARIANO MARCOS MEMORIAL STATE UNVERSITY
South La Union Campus
College of Computer Science
Consolacion, Agoo, La Union

Name: _____

Address: _____

Direction: Please read each of the items carefully. Using the scale below, please check (/) the appropriate column that best describes your answer regarding on the usability "IOT-BASED FLOOD ALERT SYSTEM FOR NARVACAN, STO. TOMAS, LA UNION".

Scale	Descriptive Equivalent Rating
7	Strongly Agree
6	Agree
5	Somewhat Agree
4	Neutral
3	Somewhat Disagree
2	Disagree
1	Strongly Disagree

System Usefulness	1	2	3	4	5	6	7	NA
1. How often do you use the system to check for flood alerts?								
2. Do you find the system helpful in making informed decisions about your safety during flood events?								
3. Does the system provide enough information to prepare for floods effectively?								

4. Has the system helped you feel more secure and prepared for flood emergencies?								
5. Do you find the system a valuable tool for managing flood risks in your community?								
6. Overall, how satisfied are you with the usefulness of the system?								
Information Quality	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**NA**
7. Do you believe the information provided by the system is accurate and reliable?								
8. Is the information provided by the system timely and up-to-date?								
9. Do you find the information provided by the system to be clear and easy to understand?								
10. Does the information the system provides address your community's specific needs?								
11. How confident are you in the accuracy of the information provided by the system?								
12. Overall, how satisfied are you with the system's quality of information?								
Interface Quality	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**NA**
13. Do you find the system easy to navigate and use?								
14. Do you find the system interface visually appealing and user-friendly?								
15. Is the system accessible to users with different levels of technical expertise?								
16. Does the system provide users with clear instructions and helpful feedback?								
17. How intuitive do you find the system's design and layout?								
18. Overall, how satisfied are you with the system interface?								

Thank you very much.

Respondent's Signature	**Date**

APPENDIX E

Matrix of Acceptability Results

1. System Usefulness

NO. RESPONDENTS	Q1	Q2	Q3	Q4	Q5	Q6
01	7	7	7	7	7	7
02	7	6	6	7	7	7
03	7	6	6	7	6	6
04	5	7	7	5	7	7
05	7	7	7	7	7	7
06	7	7	7	7	7	7
07	7	5	6	5	6	6
08	7	7	7	7	7	7
09	7	7	7	7	7	7
10	7	7	7	7	7	7
11	7	7	7	7	7	7
12	7	7	7	7	7	7
13	5	6	7	7	7	7
14	7	7	7	7	7	7
15	7	6	7	6	6	6
16	7	7	7	7	7	7
17	7	7	7	7	7	7
18	7	7	7	7	7	7
19	7	7	7	7	7	7
20	7	6	7	7	7	7
21	7	7	7	7	7	7
22	7	7	7	7	7	7
23	7	7	7	7	7	7
24	7	7	7	7	7	6
25	7	7	7	7	7	7
26	7	7	6	6	7	7
27	7	7	7	7	7	7
28	7	7	7	7	7	7
29	7	7	7	7	7	7
30	7	7	7	7	7	7
TOTAL:	**6.93**	**6.77**	**6.87**	**6.87**	**6.90**	**6.87**

Total Grand Mean: 6.87

2. Information Quality

NO. RESPONDENTS	Q1	Q2	Q3	Q4	Q5	Q6
01	7	7	7	7	7	7
02	7	6	7	7	6	7
03	7	7	6	7	6	6
04	5	7	7	7	7	7
05	7	7	7	7	7	7
06	7	7	7	7	7	7
07	6	7	6	6	5	6
08	6	7	7	7	7	7
09	7	7	7	7	7	7
10	7	7	7	7	7	7
11	7	7	7	7	7	7
12	5	6	7	5	6	7
13	7	7	7	7	7	6
14	7	6	7	7	7	7
15	6	6	7	6	6	7
16	7	7	7	7	7	7
17	7	7	7	7	7	7
18	7	7	7	7	7	7
19	7	7	6	6	7	7
20	7	6	7	6	7	6
21	7	7	7	7	7	7
22	7	7	7	7	7	7
23	7	7	7	7	7	7
24	7	7	7	7	6	7
25	7	6	7	7	7	7
26	7	7	7	7	7	7
27	7	7	7	7	7	7
28	7	7	7	7	7	7
29	7	7	7	7	7	7
30	7	7	7	7	7	7
TOTAL:	**6.83**	**6.80**	**6.90**	**6.80**	**6.77**	**6.87**

Total Grand Mean: 6.83

3. Interface Quality

NO. RESPONDENTS	Q1	Q2	Q3	Q4	Q5	Q6
01	7	7	6	7	7	6
02	7	6	6	6	7	7
03	7	7	7	7	7	7
04	5	6	7	5	6	7
05	7	7	7	7	7	7
06	7	7	7	7	7	7
07	7	7	7	7	7	7
08	7	7	7	7	7	7
09	7	7	7	7	7	7
10	6	6	6	6	7	6
11	7	7	7	7	7	7
12	7	7	7	7	7	7
13	7	7	7	7	7	7
14	7	6	6	5	6	7
15	7	7	6	7	6	7
16	7	7	7	7	7	7
17	7	7	7	7	7	7
18	7	7	7	7	7	7
19	7	6	7	7	7	7
20	6	7	6	7	7	7
21	7	6	7	6	6	7
22	7	7	7	7	7	7
23	7	7	7	7	6	7
24	7	7	7	7	7	7
25	7	7	7	7	7	7
26	7	7	7	7	7	7
27	7	7	6	7	7	6
28	7	7	7	7	7	7
29	7	7	7	7	7	7
30	7	7	7	7	7	7
TOTAL:	**6.93**	**6.77**	**6.87**	**6.87**	**6.90**	**6.87**

Total Grand Mean: 6.87

APPENDIX I

Application for Ethics Review

	Application for Ethics Review of a New Protocol

Instructions to the Researcher: Please accomplish this form and ensure that you have included in your submission the documents that you checked below (in Section 3. Checklist of Documents).

1. General Information			
*Title of Study	IOT-BASED FLOOD ALERT SYSTEM FOR NARVACAN, STO. TOMAS, LA UNION		
*RETC Code (*To be provided by RETC*)		*Study Site	Narvacan, Sto. Tomas, La Union
*Name of Researcher)	Carl Anthony V. Dela Cruz	Contact Information	*Tel No:
			*Mobile No: +63963-987-337
*Co-researcher (*if any*)	Jinky Rose L. Dela Cruz Justine O. Estolero Brenda J. Lachica Rey John E. Pineda Andrea P. Viduya		Fax No:
			*Email: cadelacruz@student.dmmmsu.edu.ph
*Institution	Don Mariano Marcos Memorial State University-South La Union Campus		
*Address of Institution	Barangay Consolacion, Agoo, La Union, 2504		
*Type of Study	☐ Clinical Trial (Sponsored)	☐ Biomedical Research (Retrospective, Prospective and diagnostic studies)	
	☐ Clinical Trials (Researcher-initiated)	☐ Stem Cell Research	
	☐ Health Operations Research (Health Programs and Policies)	☐ Genetic Research	
	☐ Public Health / Epidemiologic Research	☐ Social / Behavioral Research	
	☒ Others: Descriptive and Developmental Research		
	☐ Multicenter (International)	☐ Multicenter (National)	☐ Single Site
*Source of Funding	☒ Self-funded	☐ Sponsored by a Pharmaceutical Company	
	☐Government-Funded	Specify:	
	☐ Scholarship/Research Grant		
	☐ Institution-Funded		

APPENDIX J

Informed Consent Form (English Version)

	College of Computer Science

This informed consent form is for the barangay councils, and residents of Barangay Narvacan, Sto. Tomas, La Union who we are inviting to participate in this capstone research project, titled " IOT- BASED FLOOD ALERT SYSTEM FOR NARVACAN, STO. TOMAS, LA UNION "

Name of Principal Investigator: Carl Anthony V. Dela Cruz, Jinky Rose L. Dela Cruz, Justine O. Estolero, Brenda J. Lachica, Rey John E. Pineda, Andrea P. Viduya.

Name of Organization: Don Mariano Marcos Memorial State University – South La Union Campus

Name of Sponsor (if applicable):

Name of Project and Version: IOT- BASED FLOOD ALERT SYSTEM FOR NARVACAN, STO. TOMAS, LA UNION

Part I: Information Sheet

Introduction

Flooding is a major natural hazard in the country causing significant damage to property, infrastructure, and even people's lives. Barangay Narvacan, Sto. Tomas, La Union is vulnerable to flooding due to low-lying topography and proximity to rivers and sea. To avert such situation, it is important to monitor and receive timely alerts about the water level situation. This study aims to develop an IoT-based flood alert system for Narvacan, Sto. Tomas, La Union. Specifically, the study seeks to achieve the following objectives:1) develop a prototype of the IoT-Based Flood Alert System adopting the Agile methodology and 2) assess the system's level of acceptability using the User Technology Acceptability Questionnaire. Throughout the study, descriptive and developmental research methods will be employed to accomplish its objectives. It focuses on the water level monitoring and early warning system via SMS that alerts individuals for potential flood event, giving time to evacuate and abate impact of flooding.

Purpose of the research

The purpose of the research is to develop an IoT-based flood alert system for Narvacan, Sto. Tomas, La Union.Throughout the study, descriptive and developmental research methods will be employed to accomplish its objectives. It focuses on the water level monitoring and early warning system via SMS that alerts individuals for potential flood event, giving time to evacuate

and abate impact of flooding particularly in areas vulnerable to the effects of climate change and human activities.

Type of Research Intervention

The research intervention is related to develop a prototype of the IoT-Based Flood Alert System adopting the Agile methodology and assess the system's level of acceptability. Without additional information, it's challenging to determine the specific details of the intervention. However, it's possible that the system may involve the use of data collection tools such as decision support or satellite imagery to monitor changes in the coastal environment, including changes in water level, erosion, or sedimentation patterns. The system could also involve the development of a database or visualization tool that allows users to track and analyze these trends over time. The research intervention appears to be aimed at providing valuable information to stakeholders in the coastal areas of Narvacan Sto. Tomas, La Union, such as local governments, environmental groups, and researchers, to help them make informed decisions and plan for the future.

Participant Selection

As researchers in the field of Computer Science, we would like to invite you to participate in our research project. We believe that your expertise in evaluating model performance would be a valuable addition to our team, as we work on developing our own model.

Voluntary Participation

Your involvement in this study is entirely voluntary, and you have the freedom to choose whether or not to participate. Declining to participate will not impact the care, services, or benefits that you are entitled to receive.

Procedures

A. The researchers will be collecting relevant data on coastal trends in Narvacan, Sto. Tomas, La Union, such as historical water levels, erosion rates, sedimentation patterns, and other environmental indicators and with the proper practicing of social distancing and wearing face mask will gathering interview to the respondents.

B. In developing the system with designing and building a system to develop an IoT-based flood alert system for Narvacan, Sto. Tomas, La Union. The researchers could involve

selecting appropriate data collection tools, such as sensors or remote sensing techniques, and developing a database or visualization tool to process and display the data.

C. The researchers will be testing the system which evaluates the performance of the system by comparing its results to the baseline data and other established measures IoT-based flood alert system. As part of the health protocol in testing the researchers sanitize and observe proper disinfection.

D. The researchers will also be refining the system based on the results of the testing, making adjustments and refinements to the system to improve its accuracy and effectiveness.

E. The researchers will implement the system once the system is fully developed and refined: it can be deployed in the field to collect ongoing data on IoT-based flood alert system for Narvacan, Sto. Tomas, La Union with proper safety and health care protocols.

Duration

The duration of the research study is nine (9) months from August 2023 to April 2024.

Risks

Participants in this study will not be required to disclose any confidential information, such as their email address, password, records, birthdate, or phone number. The study will involve face-to-face testing, and both the researchers and participants will adhere to minimum health protocols and standards. The equipment that participants will use during testing will also be sanitized to ensure safety.

Benefits

Taking part in this research will be highly beneficial in evaluating the experiment's outcomes. Participation is expected to assist the researchers in gaining a deeper understanding of the factors that significantly influence the study, leading to improved results.

Reimbursements

Participants will not be charged for participating in this research. However, their time and work will be greatly appreciated.

Confidentiality

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All data collected as part of this study that could potentially identify participants will be kept confidential and shared only with their consent. Participation in the study may draw attention from other parties interested in the project, but the researcher will not share any personal information of participants with anyone outside the research team. The researcher will maintain strict confidentiality of all information gathered, with access to the data limited only to authorized members of the research team. No confidential protocol information will be made public in this research.

Sharing the Results

The researchers will ensure that any information collected as part of the study that could be used to identify participants will be kept confidential and shared only with their consent. Personal data will only be accessible to authorized members of the research team, and all collected data will be treated with strict confidentiality. The researchers will follow the prohibition of disclosing details about the study technique. Participants' information will be accessible only to the research team, and no one outside of the team will have access to it.

Right to Refuse or Withdraw

Participants have the right to withdraw their participation at any time without losing any legal claims, rights, or remedies. They are not obligated to participate and may decline to do so at any point. Participation in the research project is voluntary, and participants may choose not to answer survey questions without any negative consequences. They are not under any pressure to give their permission and can decide not to participate if they do not want to.

Who to Contact

*This proposal has been reviewed and approved by **DMMMSU Research Ethics Committee**, which is a committee whose task it is to make sure that research participants are protected from harm. If you wish to find about more about the Committee, contact:*

Joel C. Estacio

Chair, DMMMSU REC

reco.dmmmsu.edu.ph

	College of Computer Science

Do you know that you do not have to take part in this study if you do not wish to? You can say No if you wish to? Do you know that you can ask me questions later, if you wish to? Do you know that I have given the contact details of the person who can give you more information about the study?

You can ask me any more questions about any part of the research study, if you wish to. Do you have any questions? You can contact me through the following:

Dela Cruz, Carl Anthony V.
Group Leader,
Email: cadelacruz@student.dmmmsu.edu.ph
Contact Number: 09639874337

Part II: Certificate of Consent

I have read the foregoing information, or it has been read to me. I have had the opportunity to ask questions about it and any questions I have been asked have been answered to my satisfaction. I consent voluntarily to be a participant in this study

Name of Participant: _____

Signature of Participant: _____

Date: _____

Day/month/year

Statement by the researcher/person taking consent

I have accurately read out the information sheet to the potential participant, and to the best of my ability made sure that the participant understands that the following will be done:

1. _____

2. _____

	College of Computer Science

3. _____

I confirm that the participant was given an opportunity to ask questions about the study, and all the questions asked by the participant have been answered correctly and to the best of my ability. I confirm that the individual has not been coerced into giving consent, and the consent has been given freely and voluntarily.

A copy of this ICF has been provided to the participant.

Name of Researcher/person taking the consent _____

Signature of Researcher /person taking the consent _____

Date _____

Day/month/year

APPENDIX K

Informed Consent Form (Filipino Version)

	College of Computer Science

Ihahandog sa inyo ang pahayag ng pahintulot para sa mga konsehо at residente sa Barangay Narvacan, Sto. Tomas, La Union na iniimbitahan naming makilahok sa proyektong pananaliksik na ito, na may pamagat na "IOT- BASED FLOOD ALERT SYSTEM PARA SA NARVACAN, STO. TOMAS, LA UNION."

Pangunahing Investigador: Carl Anthony V. Dela Cruz, Jinky Rose L. Dela Cruz, Justine O. Estolero, Brenda J. Lachica, Rey John E. Pineda, Andrea P. Viduya.

Pangalan ng Organisasyon: Don Mariano Marcos Memorial State University – South La Union Campus

Pangalan ng Sponsor (kung mayroon):

Pangalan ng Proyekto at Bersyon: IOT- BASED FLOOD ALERT SYSTEM PARA SA NARVACAN, STO. TOMAS, LA UNION

Bahagi I: Impormasyon Sheet

Introduksyon

Ang pagbaha ay isang pangunahing natural na panganib sa bansa na nagdudulot ng malaking pinsala sa ari-arian, imprastruktura, at maging sa buhay ng mga tao. Ang Barangay Narvacan, Sto. Tomas, La Union ay maaaring mapinsala sa pagbaha dahil sa mababang topograpiya at malapit sa mga ilog at dagat. Upang maiwasan ang ganitong sitwasyon, mahalaga na bantayan at makatanggap ng timely na mga abiso tungkol sa kalagayan ng antas ng tubig. Layunin ng pag-aaral na ito na mag-develop ng isang IoT-based flood alert system para sa Narvacan, Sto. Tomas, La Union. Partikular, ang pag-aaral ay naglalayon na makamit ang mga sumusunod na layunin: 1) mag-develop ng isang prototype ng IoT-Based Flood Alert System na gumagamit ng Agile methodology at 2) suriin ang antas ng pagtanggap ng sistema gamit ang User Technology Acceptability Questionnaire. Sa buong pag-aaral, gagamitin ang mga deskriptibo at developmental na mga pamamaraan ng pananaliksik upang makamit ang mga layunin nito. Nakatuon ito sa pagmamanman ng antas ng tubig at sistema ng maagang babala sa pamamagitan ng SMS na nag-aabiso sa mga indibidwal para sa potensyal na pangyayaring baha, na nagbibigay ng oras upang lumikas at bawasan ang epekto ng pagbaha.

Layunin ng Pananaliksik

Ang layunin ng pananaliksik ay mag-develop ng isang IoT-based flood alert system para sa Narvacan, Sto. Tomas, La Union. Sa buong pag-aaral, gagamitin ang mga deskriptibo at developmental na mga pamamaraan ng pananaliksik upang makamit ang mga layunin nito. Nakatuon ito sa pagmamanman ng antas ng tubig at sistema ng maagang babala sa pamamagitan ng SMS na nag-aabiso sa mga indibidwal para sa potensyal na pangyayaring baha, na nagbibigay ng oras upang lumikas at bawasan ang epekto ng pagbaha lalo na sa mga lugar na vulnerable sa epekto ng pagbabago ng klima at gawain ng tao.

Uri ng Pananaliksik na Intervensyon

Ang interbensyon sa pananaliksik ay may kinalaman sa pag-develop ng isang prototype ng IoT-Based Flood Alert System na gumagamit ng Agile methodology at suriin ang antas ng pagtanggap ng sistema. Nang walang karagdagang impormasyon, mahirap malaman ang mga partikular na detalye ng interbensyon. Gayunpaman, posible na ang sistema ay kasama ang paggamit ng mga tool sa pagtitipon ng datos tulad ng desisyon na suporta o mga larawang satellite upang

bantayan ang mga pagbabago sa kapaligiran sa baybayin, kasama ang mga pagbabago sa antas ng tubig, erosion, o mga pattern ng sedimento. Maaari ring isama ng sistema ang pag-develop ng isang database o tool sa pagpapakita na nagpapahintulot sa mga tagagamit na magsubaybay at mag-analisa ng mga trend na ito sa paglipas ng panahon. Ang interbensyon sa pananaliksik ay tila naglalayong magbigay ng mahalagang impormasyon sa mga stakeholder sa mga baybayin ng Narvacan Sto. Tomas, La Union, tulad ng lokal na pamahalaan, mga grupo sa kalikasan, at mga mananaliksik, upang tulungan silang gumawa ng mga maunawaang desisyon at magplano para sa hinaharap.

Pagpili ng mga Kalahok

Bilang mga mananaliksik sa larangan ng Computer Science, nais naming imbitahan ka na makilahok sa aming proyektong pananaliksik. Naniniwala kami na ang iyong kaalaman sa pagtatasa ng pagganap ng modelo ay magiging mahalagang dagdag sa aming koponan, habang gumagawa kami ng aming sariling modelo.

Boluntaryong Pakikilahok

Ang iyong paglahok sa pag-aaral na ito ay lubos na boluntaryo, at may kalayaan kang pumili kung nais mong makilahok o hindi. Ang pagtanggi na makilahok ay hindi makakaapekto sa pangangalaga, serbisyo, o mga benepisyo na iyong karapat-dapat na matanggap.

Proseso

A. Ang mga mananaliksik ay magkokolekta ng kaukulang datos sa mga trend sa baybayin sa Narvacan Sto. Tomas, La Union, tulad ng mga nakaraang antas ng tubig, rate ng erosion, mga pattern ng sedimento, at iba pang mga indikador sa kapaligiran.

B. Sa pag-develop ng sistema, kasama ang pagdidisenyo at pagtatayo ng isang sistema para sa pag-develop ng isang IoT-based flood alert system para sa Narvacan, Sto. Tomas, La Union. Maaaring isama ng mga mananaliksik ang pagpili ng angkop na mga tool sa pagtitipon ng datos, tulad ng mga sensor o mga teknik sa malayong pag-aaral, at pag-develop ng isang database o tool sa pag-visualize upang iproseso at ipakita ang mga datos.

C. Ang mga mananaliksik ay magte-test ng sistema na nagtatasa ng pagganap ng sistema sa pamamagitan ng paghahambing ng mga resulta nito sa batayan na datos at iba pang mga nakatalagang sukat ng IoT-based flood alert system.

D. Ang mga mananaliksik ay magrerefine din ng sistema batay sa mga resulta ng testing, gagawa ng mga pag-aayos at pagpapainam sa sistema upang mapabuti ang kanyang katiyakan at kahusayan.

E. Ang mga mananaliksik ay ipapatupad ang sistema kapag ang sistema ay ganap na nadevelop at na-refine; maaari itong ilunsad sa field upang magkolekta ng patuloy na datos sa IoT-based flood alert system para sa Narvacan, Sto. Tomas, La Union.

Tagal

Ang tagal ng pag-aaral ng pananaliksik ay limang (5) buwan.

Panganib

Ang mga kalahok sa pag-aaral na ito ay hindi hinihilingang ibunyag ang anumang kumpidensyal na impormasyon, tulad ng kanilang email address, password, rekord, petsa ng kapanganakan, o numero ng telepono. Ang pag-aaral ay magpapakita ng face-to-face testing, at parehong ang mga mananaliksik at mga kalahok ay susunod sa minimum na mga protocol at pamantayan sa kalusugan. Ang kagamitan na gagamitin ng mga kalahok sa testing ay magiging sanitized din upang siguraduhing ligtas.

Benepisyo

Ang paglahok sa pananaliksik na ito ay lubhang makakatulong sa pagtatasa ng mga resulta ng eksperimento. Inaasahan na ang paglahok ay makakatulong sa mga mananaliksik na magkaroon ng mas malalim na pang-unawa sa mga salik na lubos na nakakaapekto sa pag-aaral, na magdadala sa mas mahusay na mga resulta.

Reimbursyo

Hindi sisingilin ang mga kalahok para sa paglahok sa pananaliksik na ito. Gayunpaman, ang kanilang oras at trabaho ay lubos na pinahahalagahan.

Pagiging Kumpidensyal

Ang lahat ng datos na kolektado bilang bahagi ng pag-aaral na ito na maaaring potensyal na makakakilala sa mga kalahok ay mananatiling kumpidensyal at ibabahagi lamang sa kanilang pahintulot. Ang paglahok sa pag-aaral ay maaaring magdulot ng atensyon mula sa iba pang mga partido na interesado sa proyekto, ngunit hindi ibabahagi ng mananaliksik ang anumang personal na impormasyon ng mga kalahok sa sinumang labas ng koponan ng pananaliksik. Panatilihin ng mananaliksik ang striktong kumpidensyalidad ng lahat ng impormasyong nakuha, na may access sa mga datos limitado lamang sa mga awtorisadong miyembro ng koponan ng pananaliksik. Walang kumpidensyal na impormasyon ng protocol na ibibigay sa publiko sa pananaliksik na ito.

Pagbabahagi ng Mga Resulta

Tiyakin ng mga mananaliksik na ang anumang impormasyong nakuha bilang bahagi ng pag-aaral na maaaring gamitin upang makilala ang mga kalahok ay mananatiling kumpidensyal at ibabahagi lamang sa kanilang pahintulot. Ang personal na data ay magiging accessible lamang sa mga awtorisadong miyembro ng koponan ng pananaliksik, at ang lahat ng kolektadong datos ay trinato nang may mahigpit na kumpidensyalidad. Susundan ng mga mananaliksik ang ipinagbabawal na pagbibigay ng mga detalye tungkol sa teknik ng pag-aaral. Ang impormasyon ng mga kalahok ay magiging accessible lamang sa koponan ng pananaliksik, at walang sinuman sa labas ng koponan ang magkakaroon ng access dito.

Karapatan na Tumanggi o Magwithdraw

May karapatan ang mga kalahok na magwithdraw ng kanilang partisipasyon anumang oras nang hindi nawawala ang anumang legal na claim, karapatan, o remedyo. Hindi sila obligado na makilahok at maaari silang tumanggi anumang oras. Ang paglahok sa proyektong pananaliksik ay boluntaryo, at maaaring piliin ng mga kalahok na hindi sagutin ang mga tanong sa survey nang walang anumang negatibong kahihinatnan. Hindi sila pinipilit na magbigay ng kanilang pahintulot at maaari silang magpasya na hindi makilahok kung hindi nila nais.

	College of Computer Science

Sino ang Kukontakin

Ang proposal na ito ay na-review at na-aprubahan ng DMMMSU Research Ethics Committee, na isang komite na ang tungkulin ay tiyakin na ang mga kalahok sa pananaliksik ay protektado mula sa pinsala. Kung nais mong malaman ang higit pa tungkol sa Komite, makipag-ugnayan kay:

Joel C. Estacio

Chair, DMMMSU REC

rec@dmmmsu.edu.ph

Alam mo ba na hindi mo kailangang makilahok sa pag-aaral na ito kung hindi mo nais? Maaari kang mag-sabi ng Hindi kung nais mo? Alam mo ba na maaari mong ako tanungin ng mga tanong mamaya, kung nais mo? Alam mo ba na ibinigay ko ang mga detalye ng kontak ng taong maaaring magbigay sa iyo ng karagdagang impormasyon tungkol sa pag-aaral?

Maaari mong itanong sa akin ang anumang iba pang mga tanong tungkol sa anumang bahagi ng pag-aaral na pananaliksik, kung nais mo? Mayroon ka bang anumang mga tanong? Maaari mong makipag-ugnayan sa akin sa mga sumusunod:

Dela Cruz, Carl Anthony V.
Tagapamahala ng Grupo,
Email: cadelacruz@student.dmmmsu.edu.ph
Numero: 09639874337

Bahagi II: Sertipiko ng Pahintulot

Aking binasa ang nasabing impormasyon, o ito ay binasa sa akin. Aking nagkaroon ng pagkakataon na magtanong tungkol dito at anumang mga tanong na naitanong ko ay nasagot sa aking kasiyahan. Ako ay kusang pumapayag na maging isang kalahok sa pag-aaral na ito.

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Pangalan ng Kalahok: _____

Lagda ng Kahalok: _____

Petsa: _____

Araw/Buwan/Taon

Pahayag ng mananaliksik/personal na kumukuha ng pahintulot

Aking na-basa ng tama ang impormasyon sheet sa potensyal na kalahok, at sa abot ng aking kakayahan na ginawa ang mga ito upang tiyakin na ang kalahok ay nauunawaan na ang mga sumusunod ang mangyayari:

1. _____
2. _____
3. _____

Kinukumpirma ko na binigyan ng pagkakataon ang kalahok na magtanong tungkol sa pag-aaral, at ang lahat ng mga tanong na itanong ng kalahok ay nasagot ng tama at sa abot ng aking kakayahan. Kinukumpirma ko na ang indibidwal ay hindi napilitang magbigay ng pahintulot, at ang pahintulot ay ibinigay ng malaya at kusang-loob. Isang kopya ng ICF na ito ay ibinigay sa kalahok.

Isang kopya ng ICF na ito ay ibinigay sa kalahok.

Pangalan ng mananaliksik/ _____

personal na kumukuha ng pahintulot

	College of Computer Science

Lagda ng mananaliksik/ _____

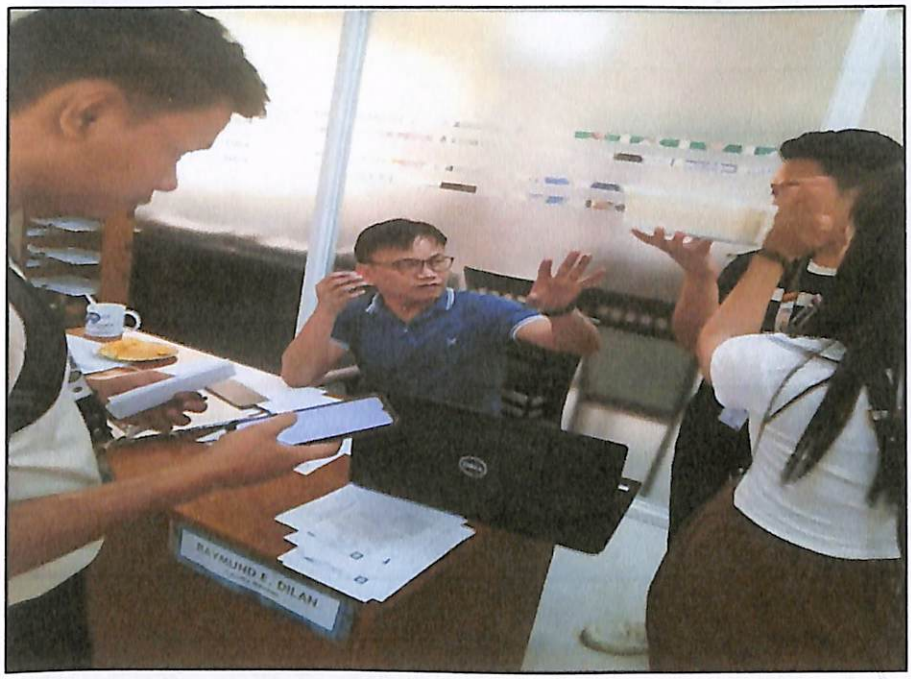
personal na kumukuha ng pahintulot

Petsa _____

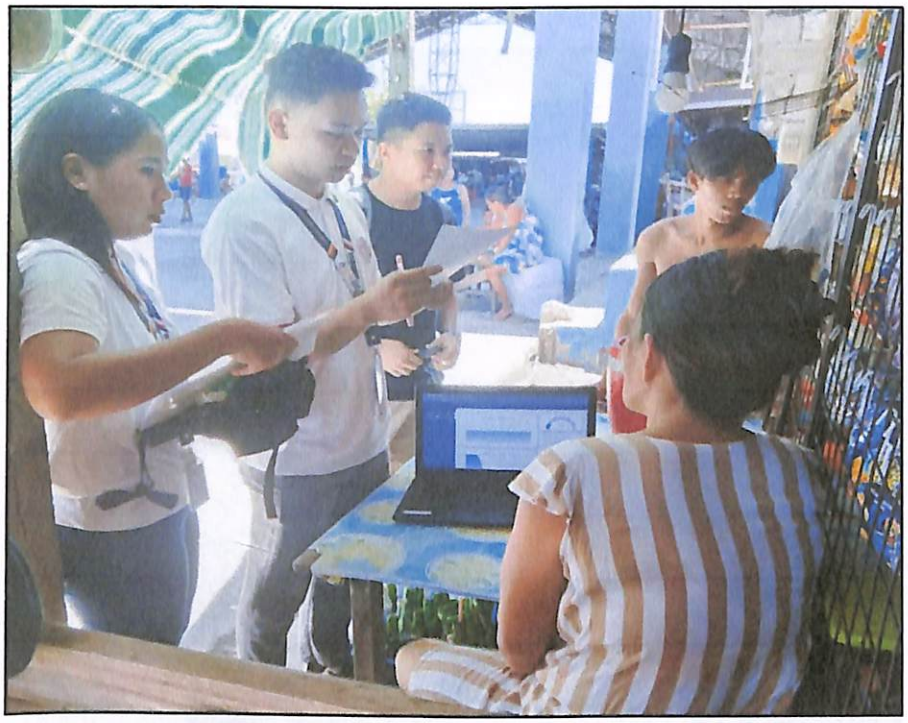
Araw/Buwan/Taon

APPENDIX M

Pictorials



The researchers conducted beta testing with 2 IT experts from the College of Computer Science.



The researchers conducted beta testing and surveyed barangay official councils and residents in Narvacan Sto. Tomas, La Union.

The researchers conducted beta testing in the office of MDRRMO at the Municipality of Santo Tomas.

APPENDIX N

Source Code

```
#include <SoftwareSerial.h>

SoftwareSerial SIM900A(10, 11);  // RX, TX

const int trigPin = 7;
const int echoPin = 6;

const int numPhoneNumbers = 4;
const char* phoneNumbers[] = { "+639915867972", "+639513083614",
"+639639874337", "+639385367087" };

// Timing constants

// Variables to control SMS sending for each distance trigger
bool canSendSMSLevel1 = true;
bool canSendSMSLevel2 = true;
bool canSendSMSLevel3 = true;

void setup() {
  SIM900A.begin(9600);
  Serial.begin(9600);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);

  Serial.println("System ready");
}

void loop() {
  sendordata();
}

void sendordata() {
  long duration, distance;

  // Trigger the ultrasonic sensor
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
```

```
  digitalWrite(trigPin, LOW);

  // Read the echo pulse
  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.034 / 2;  // Calculate distance in centimeters

  // Check distance and take action
  if (distance >= 30 && distance <= 31) {
   if (canSendSMSLevel1) {
    sendTextAlert("Alerto ti Emergency!(Level 1) Ti lebel ti danum ket dumanon iti 30-
32cm.");
    canSendSMSLevel1 = false;
   } else {
    Serial.print("Level 1 SMS is already triggered");
   }
  }

  if (distance >= 25 && distance <= 26) {
   if (canSendSMSLevel2) {
    sendTextAlert("Alerto ti Emergency!(Level 2) Maipakaammo iti bumarangy nga
umakar ti nangato a lugar.");
    canSendSMSLevel2 = false;
   } else {
    Serial.print("Level 2 SMS is alredy triggered ");
   }
  }

  if (distance >= 20 && distance <= 21) {
   if (canSendSMSLevel3) {
    sendTextAlert("Alerto ti Emergency!(Level 3) Pwersaan nga agbawit dagiti apektado
nga bumarangay iti naidisignar a balay pagbakwisan.");
    canSendSMSLevel3 = false;
   } else {
    Serial.print("Level 3 SMS is already triggered");
   }
  }

  // Re-enable SMS sending after cooldown period for each trigger
  if (!canSendSMSLevel1  && distance <= 10) {
   canSendSMSLevel1 = true;
   Serial.println("Level 1 SMS is re-enabled.");

  }
  if (!canSendSMSLevel2 && distance <= 10) {
```

```
    canSendSMSLevel2 = true;
    Serial.println("Level 2 SMS is re-enabled.");
   }
   if (!canSendSMSLevel3 && distance <= 10) {
    canSendSMSLevel3 = true;
    Serial.println("Level 3 SMS is re-enabled.");
   }

   Serial.print("Water Level: ");
   Serial.print(distance);
   Serial.println();
   delay(400);
}

void sendTextAlert(const char* message) {
  for (int i = 0; i < numPhoneNumbers; i++) {
    Serial.println("Sending text message");
    SIM900A.println("AT+CMGF=1");  // Sets the GSM Module in Text Mode
    delay(1000);
    SIM900A.println("AT+CMGS=\"" + String(phoneNumbers[i]) + "\"\r");  // Set
recipient number
    delay(2000);
    SIM900A.println(message);  // Message content
    delay(100);
    SIM900A.println((char)26);  // ASCII code of CTRL+Z
    delay(1000);
    Serial.println("Message has been sent");
    delay(2000);
  }
}



#define BLYNK_TEMPLATE_ID "TMPL69XWoP0Eb"
#define BLYNK_TEMPLATE_NAME "Water level"
#define BLYNK_AUTH_TOKEN "UOzMtz99b6Mp-Km1YlKzrJTYSPWcf49D"

/* Comment this out to disable prints and save space */
#define BLYNK_PRINT Serial


#include <ESP8266_Lib.h>
#include <BlynkSimpleShieldEsp8266.h>
```

```
// Your WiFi credentials.
// Set password to "" for open networks.
char ssid[] = "Argylle";
char pass[] = "DuinoUno";

// or Software Serial on Uno, Nano...
#include <SoftwareSerial.h>
SoftwareSerial EspSerial(2, 3);

#define trigPin 5
#define echoPin 6
#define speakerPin 8
#define speakerPin2 9


// Your ESP8266 baud rate:
#define ESP8266_BAUD 9600
ESP8266 wifi(&EspSerial);
BlynkTimer timer;


long duration;
int distance;
int percent;
int lvl;

bool isDebounced = false;

void myTimer() {
 // This function describes what will happen with each timer tick
 // e.g. writing sensor value to datastream V5
 Blynk.virtualWrite(V0, distance);
 Blynk.virtualWrite(V1, percent);
 Blynk.virtualWrite(V2, lvl);
}
void setup() {

 Serial.begin(9600);
 EspSerial.begin(ESP8266_BAUD);
 delay(100);

 pinMode(trigPin, OUTPUT);
 pinMode(echoPin, INPUT);
```

```
  pinMode(speakerPin, OUTPUT);
  pinMode(speakerPin2, OUTPUT);

  Blynk.begin(BLYNK_AUTH_TOKEN, wifi, ssid, pass, "blynk.cloud", 80);
  timer.setInterval(1000L, myTimer);
}

void loop() {
  Blynk.run();
  sensordata();
  timer.run();
  if (lvl == 1 || lvl == 2 || lvl == 3) {
    delay(10000);
  } // Reset debounce flag
  lvl = 0;

  isDebounced = false;
}
void sensordata() {

  if (!isDebounced) {
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);

    duration = pulseIn(echoPin, HIGH);
    distance = duration * 0.034 / 2;

    percent = map(distance, 40, 0, 0, 100);
    percent = constrain(percent, 0, 100);


    if (distance >= 30 && distance <= 32) {
      lvl = 1;
      tone(speakerPin, 1000);  // Play a 3kHz tone
      delay(3000);             // Play for 1 second
      noTone(speakerPin);
      delay(3000);

    } else if (distance >= 25 && distance <= 27) {
```

```
    lvl = 2;
    tone(speakerPin2, 2000);  // Play a 2kHz tone
    delay(3000);              // Play for 1 second
    noTone(speakerPin2);
    delay(3000);

  } else if (distance >= 20 && distance <= 22) {

    lvl = 3;
    tone(speakerPin, 1000);  // Play a 3kHz tone
    delay(3000);             // Play for 1 second
    noTone(speakerPin);
    delay(3000);
  }


  Serial.print("Percentage: ");
  Serial.print(percent);
  Serial.print("% Distance: ");
  Serial.print(distance);
  Serial.println();

  isDebounced = true;
  delay(300);
 }
}
```